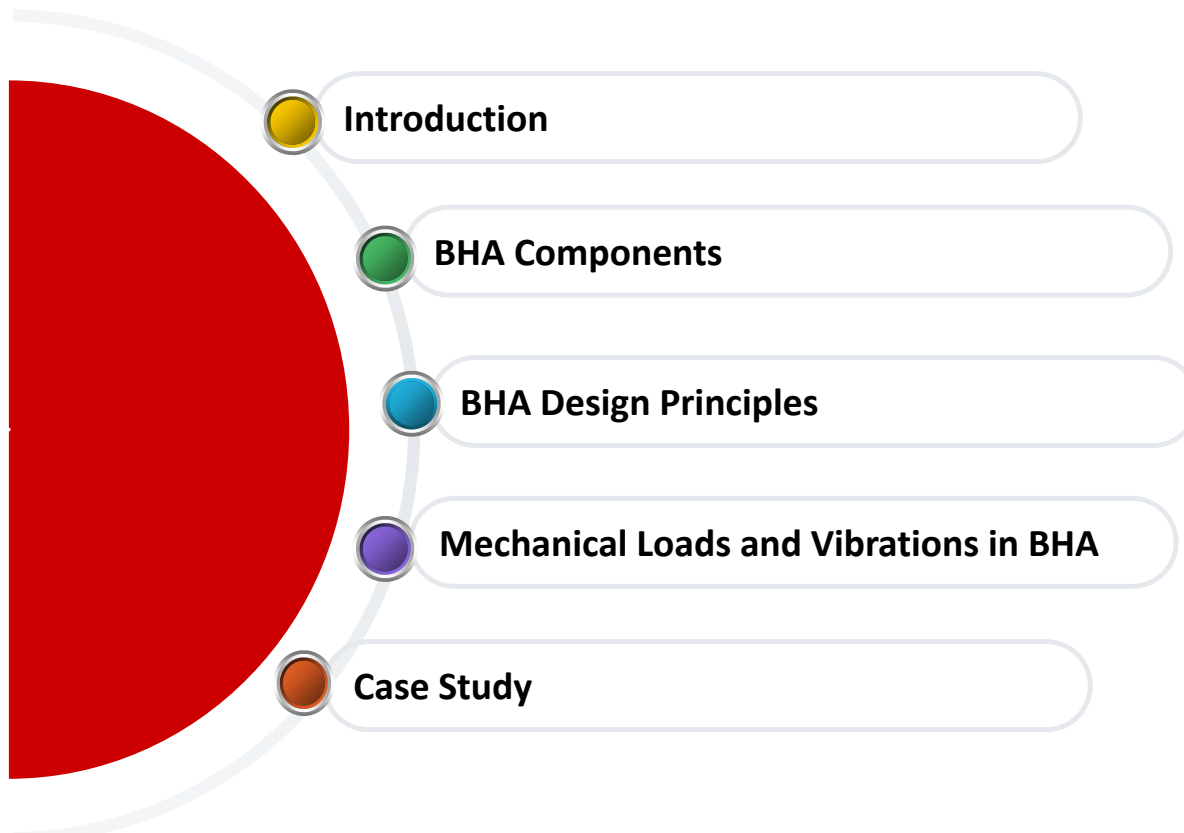




Drilling BHA

Halliburton Drilling Systems Indonesia

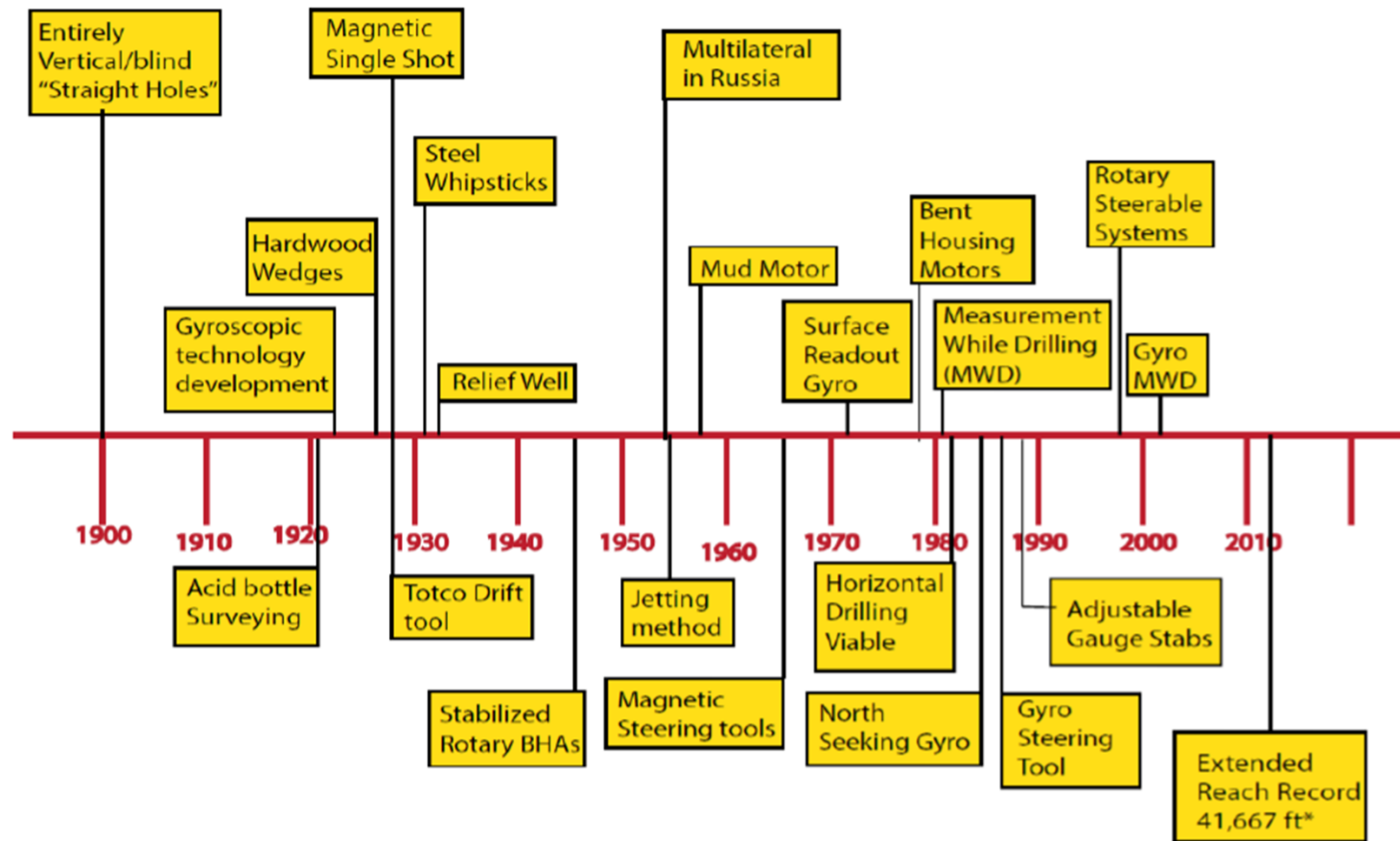
24 July 2026



History – Oil Drilling

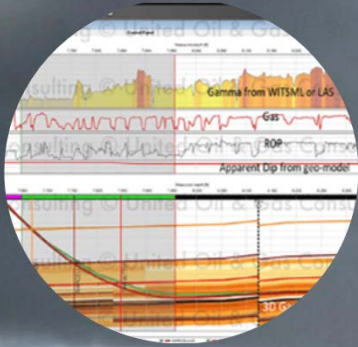
Tahun	Sejarah Perminyakan
347 SM	Pada awalnya sumur minyak yang dikenali dibor di Cina. Sumur tersebut memiliki kedalaman hingga sekitar 800 kaki (240 m) dan dibor dengan menggunakan bit yang melekat pada tiang bambu
Abad 7	Minyak bumi dikenal sebagai air terbakar di Jepang
Abad 8	Industri minyak bumi Timur Tengah
1849	Seorang insinyur Rusia F.N. Semenov menggunakan semacam kabel untuk mengebor sumur minyak di Semenanjung Absheron sepuluh tahun sebelum Kolonel Drake yang terkenal di Pennsylvania.
1859	Pemboran minyak pertama di AS dilakukan oleh William (Bill) Drake di Titusville, Pennsylvania. Ditemukan minyak dikedalaman 62.5 feet.
Akhir Abad 19	Pengeboran lepas pantai dimulai di kota Baku di lapangan Bibi-Eibat , waktu yang hampir bersamaan dengan sumur minyak pertama lepas pantai yang dibor pada tahun 1896 M di ladang Minyak Summerland Pantai California.
Sampai tahun 1970-an	Sumur minyak sebagian besar dibor secara vertikal.
1970	Technologi Directional drilling berkembang dengan inovasi teknologi tinggi.

History – Oil Drilling



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GeoSteering



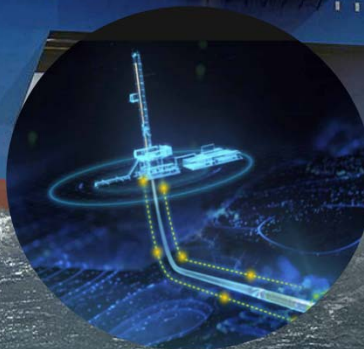
Directional Drilling



Horizontal Well



Autonomous Drilling



Casing While Drilling

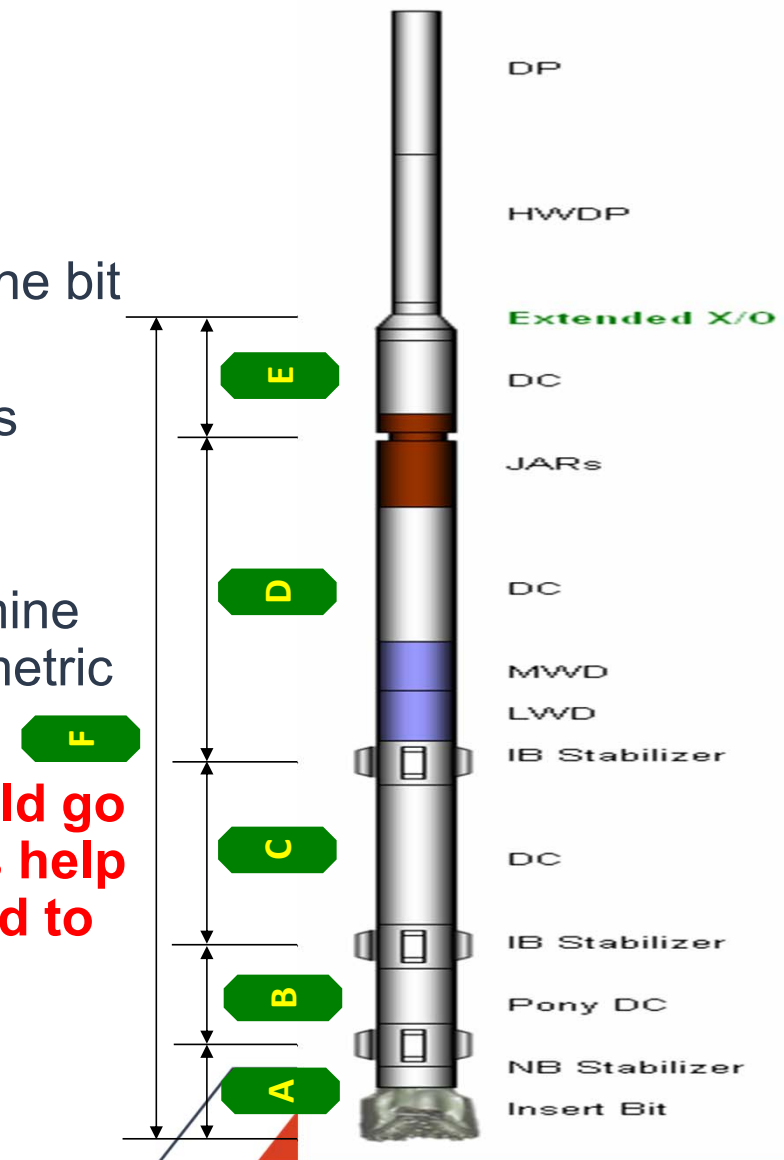


Multi Lateral Well

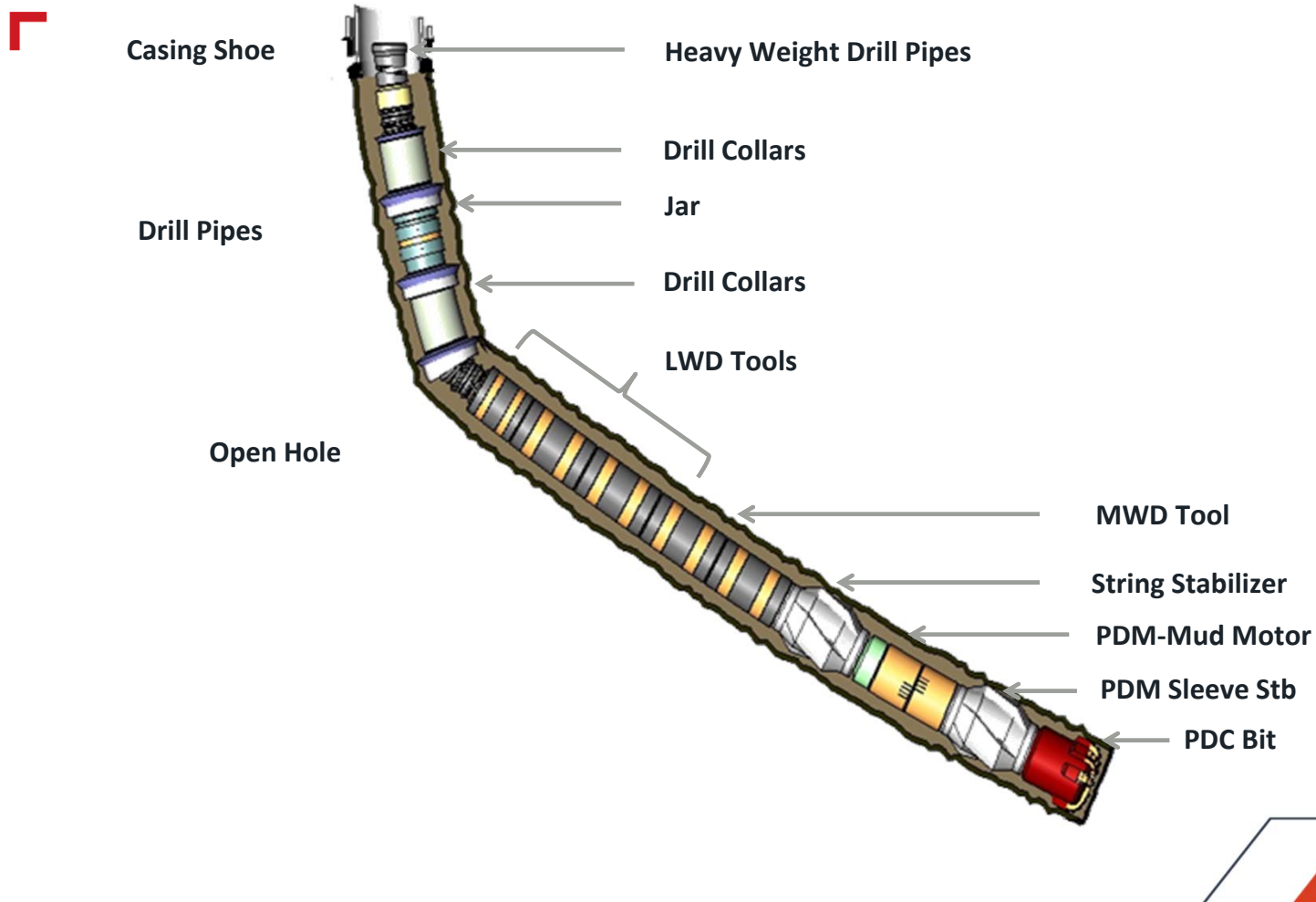


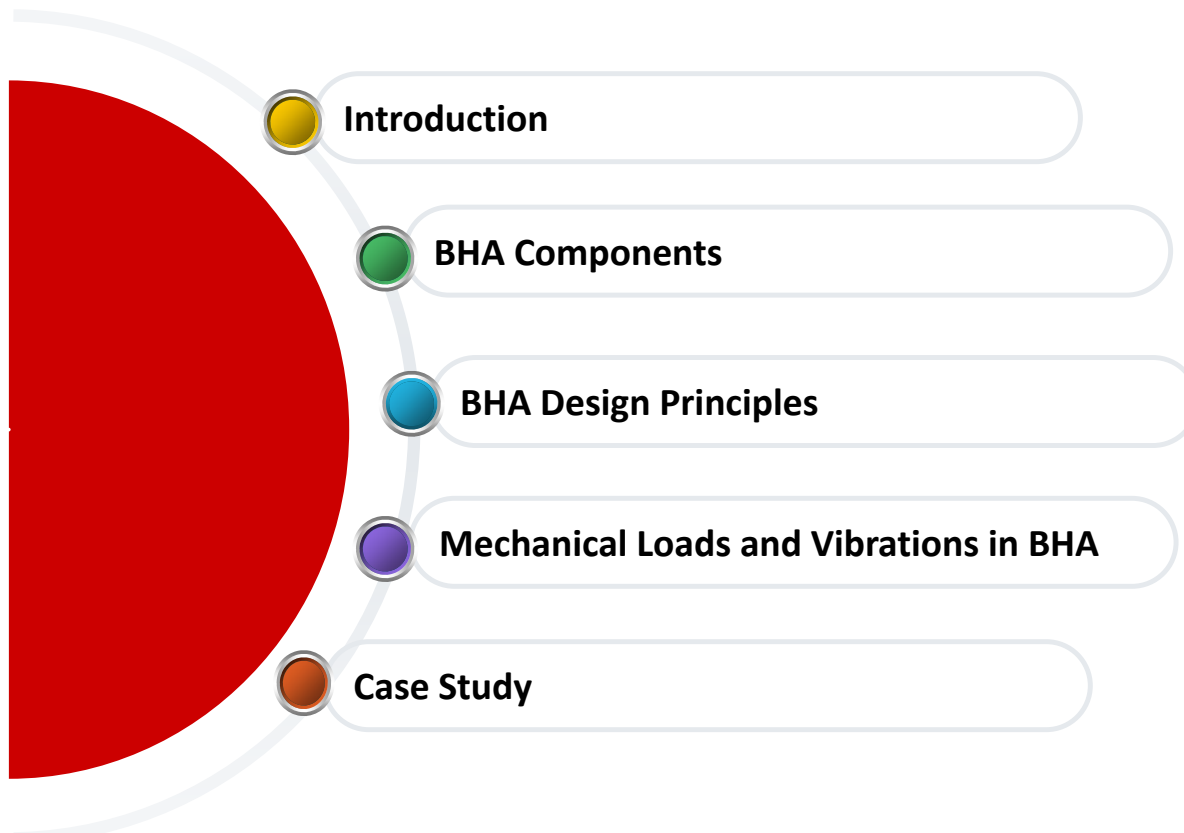
What is BHA?

- Lower part of the drill string, extending from the bit to the drill pipe.
- The assembly can consist of drill collars, subs such as stabilizers, reamers, shocks, hole-openers, and the bit sub and bit.
- The characteristics of the BHA help to determine the borehole shape, direction and other geometric characteristics.
- The **proper selection of the right BHA would go a long way in ensuring high ROP and thus help drill quickly and efficiently. This would lead to lowered drilling costs.**

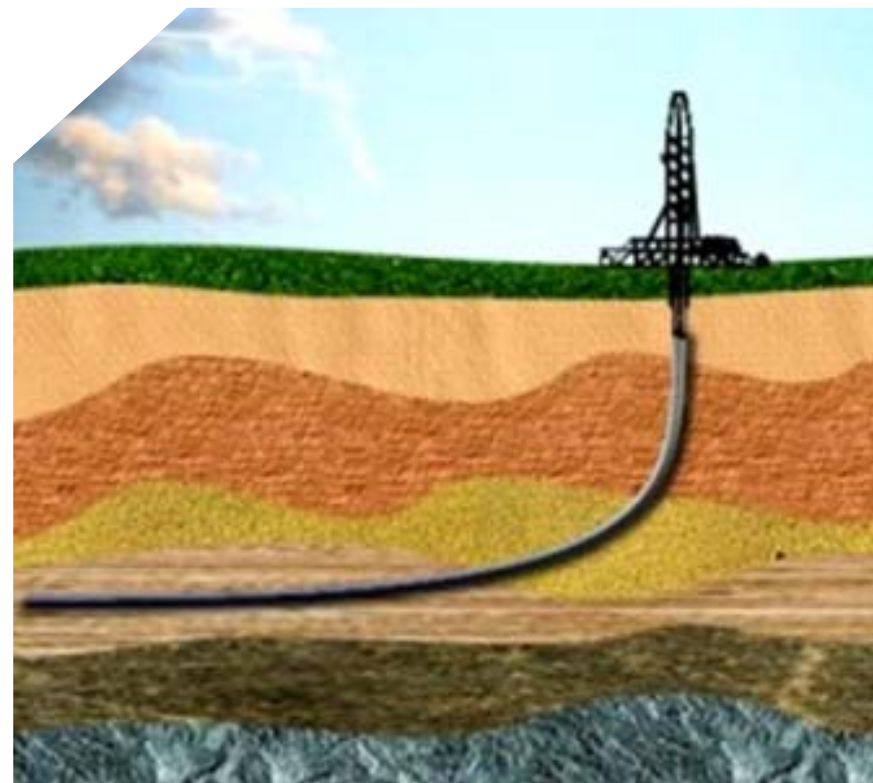
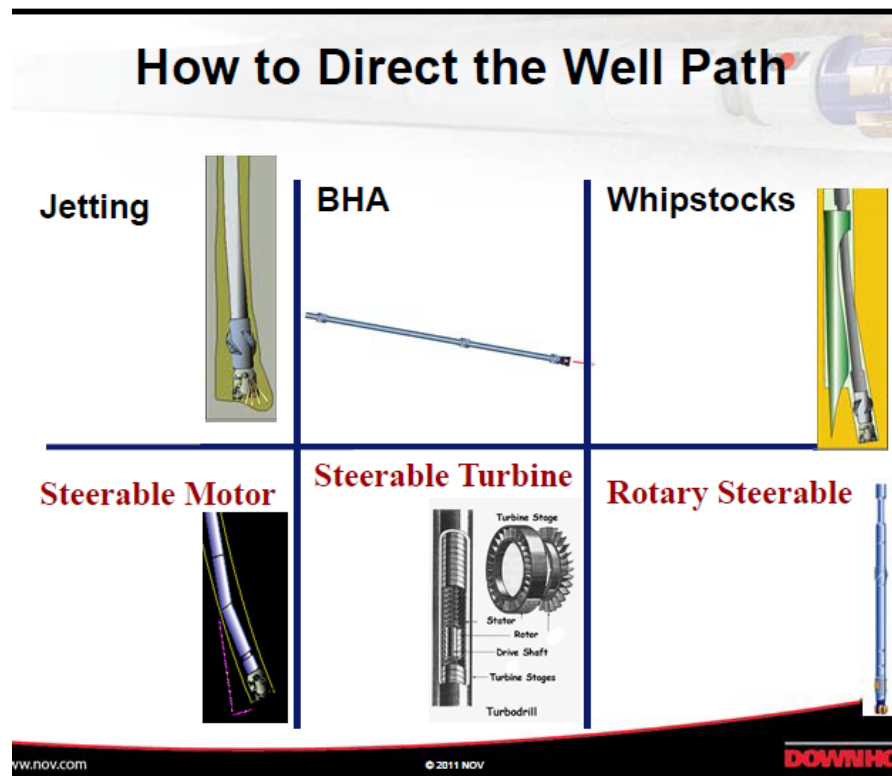


Main Components in a BHA





Drilling BHA – Deflection Tools / Methods



Mud Motor

└ Positive Displacement Mud Motors (PDM)



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Extended Learning

DOWNHOLE MOTOR

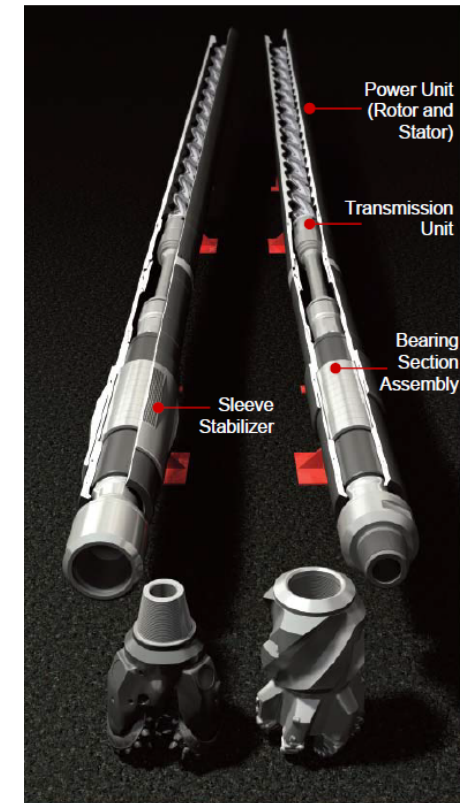
• • • •

Sliding

What Positive Displacement Mud Motor is and how it works

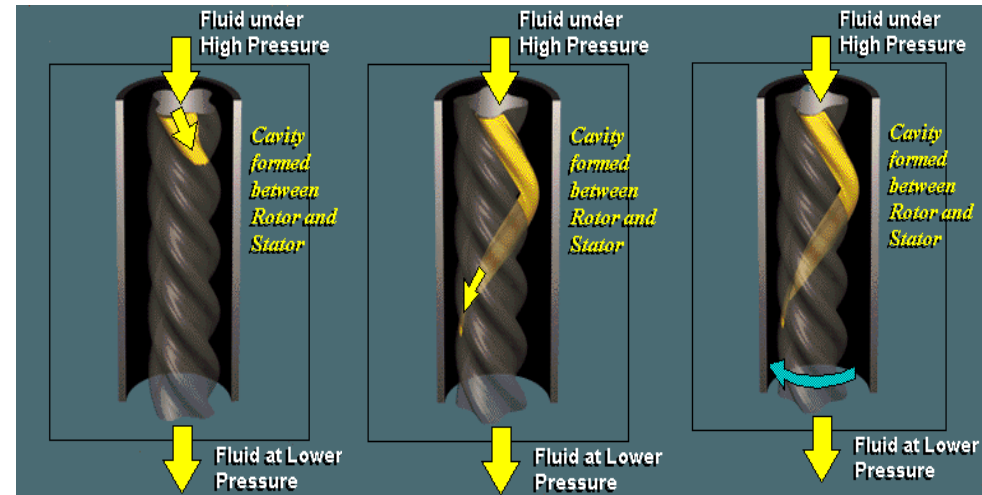


- The mud motor is a progressive cavity positive displacement pump.
- The motion of the rotor in the stator produces discrete cavities between the rotor and stator.
- The power section of a positive displacement motor (PDM) **converts the hydraulic energy of high-pressure drilling fluid to mechanical energy** in the form of torque output and RPM for the drill bit
- Additionally, to the power section, the PDM contains a transmission unit, Bearing Pack section assembly and a Bend section which are designed to provide optimum performance in all drilling application types.



Motor Power Unit (Rotor/Stator)

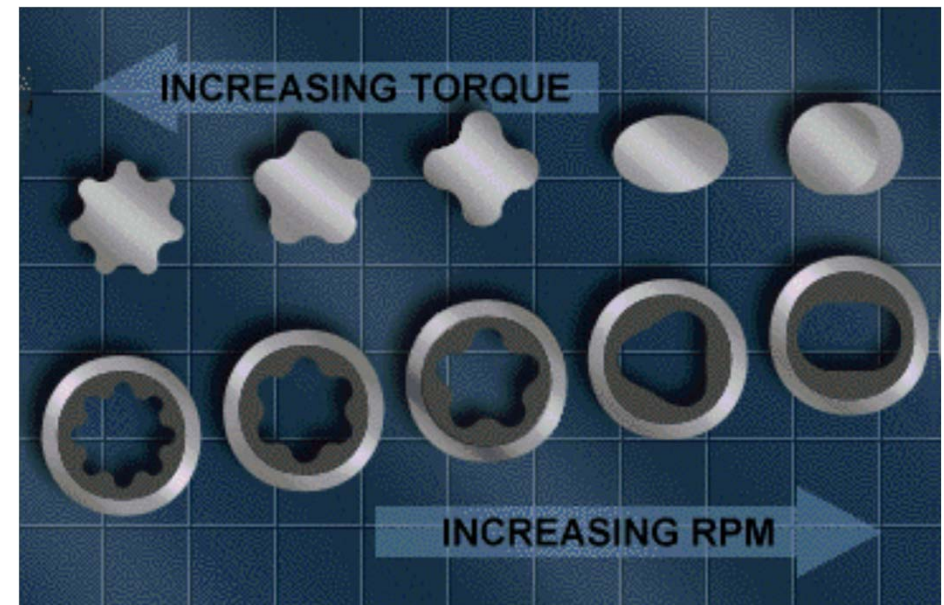
- During drilling operations, **high-pressure fluid is pumped into the top end of the power section**, where it fills the first set of open cavities.
- The differential pressure across two adjacent cavities forces the rotor to turn, and as this occurs, adjacent cavities are opened allowing the fluid to flow progressively down the length of the power section.



Motor Power Unit (Rotor/Stator)

Lobe Configuration

- The speed and torque of a power section is directly linked to the number of lobes on the rotor and stator.
- So the higher the number of lobes, the higher the torque and the lower the RPM.
- The lower the number of lobes, the lower the torque, and the higher the RPM.



Motor Power Unit (Rotor/Stator)

Factors the power section Torque and RPM

- 1. Configuration** – The number of Lobe, stage and others will determine the displacement characteristic of that design
- 2. Flow Rate** – It has a direct effect. As the flow rate increases, the RPM will increase.
- 3. Differential Pressure** – Increasing the differential pressure will increase slippage and eventually decrease RPM output.
- 4. Temperature** - High downhole temperatures affect the fit between the rotor and the elastomer (rubber) stator, resulting in increased loading of the elastomer.

Drilling Ops : Best Practice

Back-Reaming operation Recommendation

- Reduce RPM 50-70% of drilling RPM
- Reduce Flowrate 30% of Maximum Motor Specification - Recommended

TerraForce RPM limitation vs bend setting

- Reduction on the maximum RPM for 1.50° bend and avoid any rotation for 1.83° TerraForce motor.

Motor size (inches)	Motor BH (°)	Curve On bottom / Reaming (RPM)	Tangent or Vertical On Bottom / Reaming (RPM)	Circulate hole clean after TD In a Curve section / straight section (RPM)
9 5/8	<1.15	80 / 60	90 / 60	60 / 80
	1.5	30 / 30	60 / 40	40 / 60
	1.83	No Rotation	No rotation (check <u>MaxBHA</u>)	10 / 20 (Preferable, no rotation, risk of fatigue damage)
8	<1.15	80 / 60	90 / 60	60 / 90
	1.5	40 / 40	60 / 60	40 / 60
	1.83	No rotation	No rotation (check <u>MaxBHA</u>)	10 / 20 (Preferable, no rotation, risk of fatigue damage)
6 3/4 - 7	<1.15	80 / 60	90 / 60	60 / 90
	1.5	40 / 40	60 / 60	60 / 70
	1.83	No rotation	No rotation (check <u>MaxBHA</u>)	10 / 20 (Preferable, no rotation, risk of fatigue damage)
4 3/4	<1.15	80 / 60	90 / 60	60 / 90
	1.5	40 / 40	60 / 60	60 / 80
	1.83	40 / 20 (<5 DLS)	40 / 20 (check <u>MaxBHA</u>)	20 / 40 (check <u>MaxBHA</u>)

Notes

- Above table is a recommendation, but final confirmation shall come from the results of BHA modeling for Bending Moments (please refer to the well's DoS, Engineering section for Bending moment results and recommendations).
 - Motor handbook can be used as a first reference. Motor should be operated in the "Green Zone". Operating in "Ambar Zone" will apply sufficient stress in the stator connections and other components to initiate fatigue damage. Once induced, fatigue propagates at a rate determined by the number of subsequent stress cycles (RPM) and the level of stress in each cycle.
 - Once fatigue is initiated can only be repaired by machining (cutback) of the connection or scrapping the component if it is not repairable
- 1.83 BH, avoid rotation. For final hole cleaning, if well is < 35 deg, the circulation can be done with reciprocation and without rotation. For >35deg apply low RPM while reciprocating pipe.
- For BH > than 1.83 will be managed in a case-by-case basis. This type of BH shall not be rotated.
- Any deviation must be approved via MOC prior client's written agreement on covering equipment repair or replacement base on the extend of the damage.
- In case MaxBHA shows bending moments are not exceeded in any part, the drilling rotation can be applied up to 80 RPM for drilling operations (Given favorable conditions, RPM can be increased to 100).
- For Backreaming operations it is recommended to reduce RPM by 50%.

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Deflection Tools –Rotary Steerable System

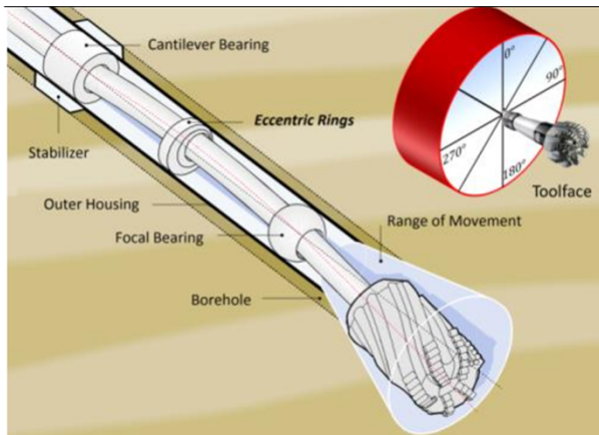
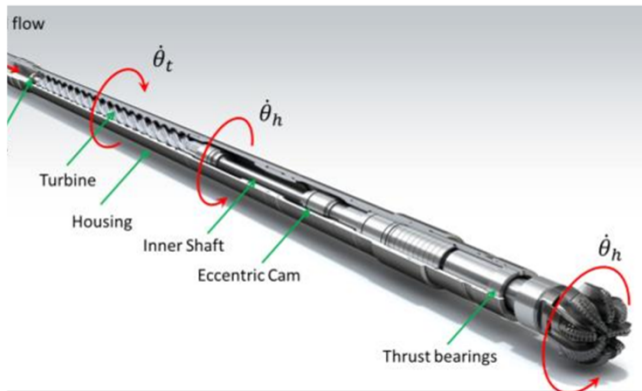


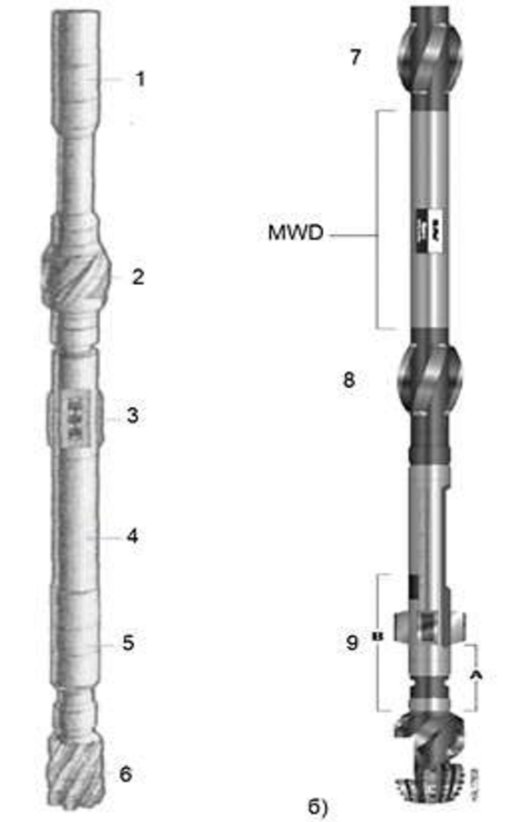
Figure 1: A typical point-the-bit RSS



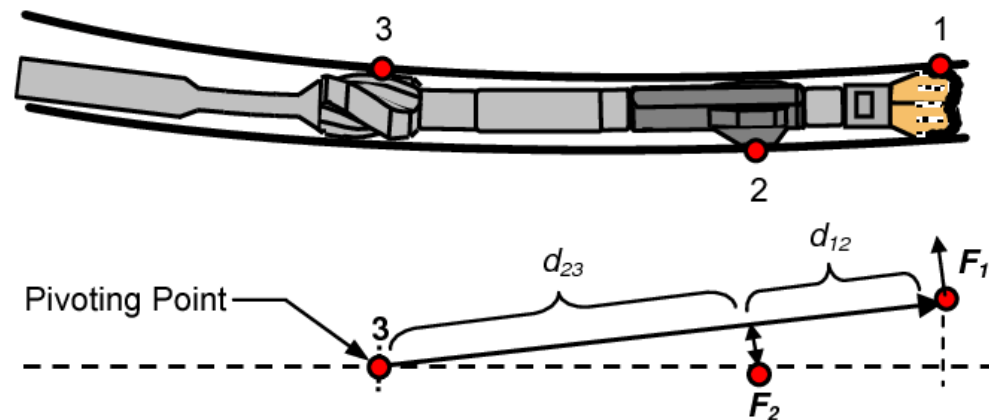
Push The Bit

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Rotary Steerable BHA

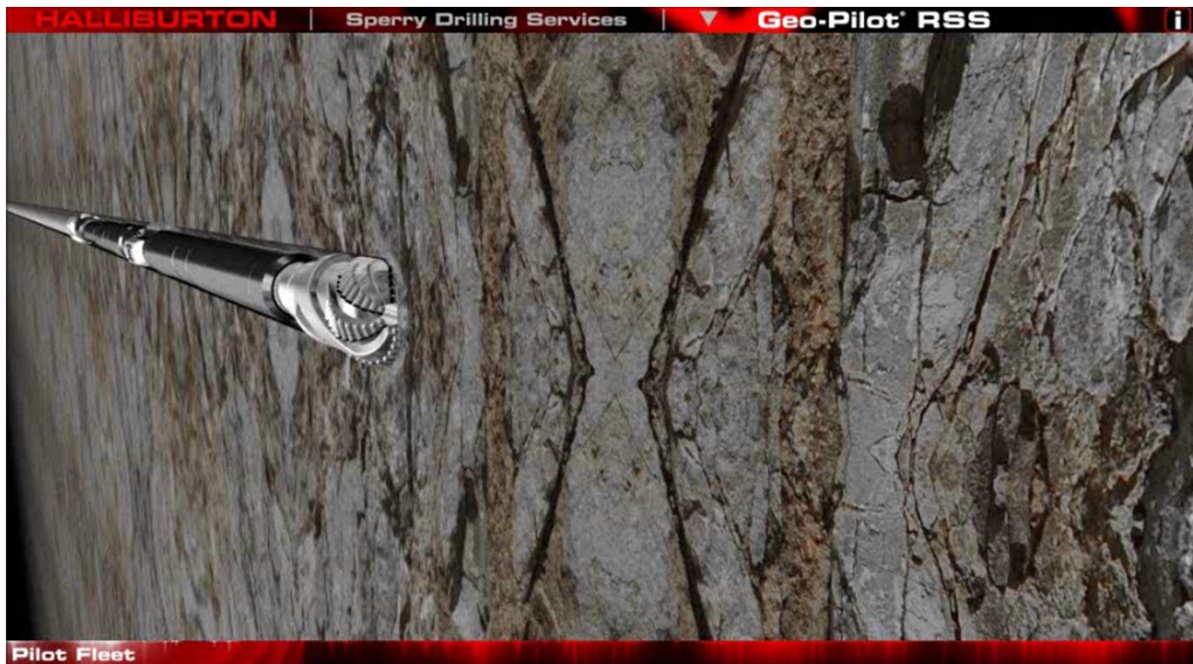


- Packed assembly to stabilize the M/LWD Collar.
- Smoother wellbore compare to mud motor.
- Reduce static pipe condition (always rotate).
- At-bit (near bit) Inclination and GR to allow precise wellbore placement.
- Two main deflecting method (point the bit and push the bit), forcing either bit (through inner shaft) or near bit stabilizer/pad.



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Rotary Steerable Principal of Operation

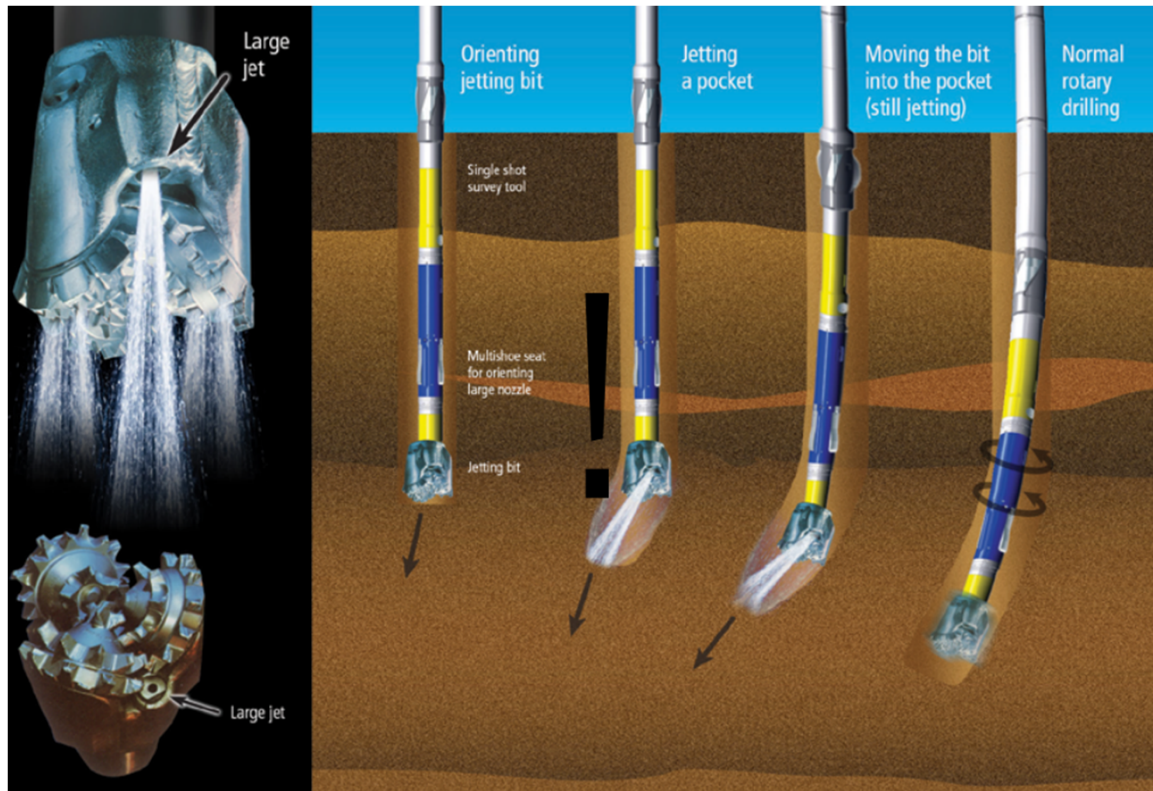


RSS Benefits

- Variable deflection for precise control
- Azimuthal gamma and inclination (ABI) 5' from the bit for precise geo-steering
- Cruise control/Vertical cruise/Vertical kickoff

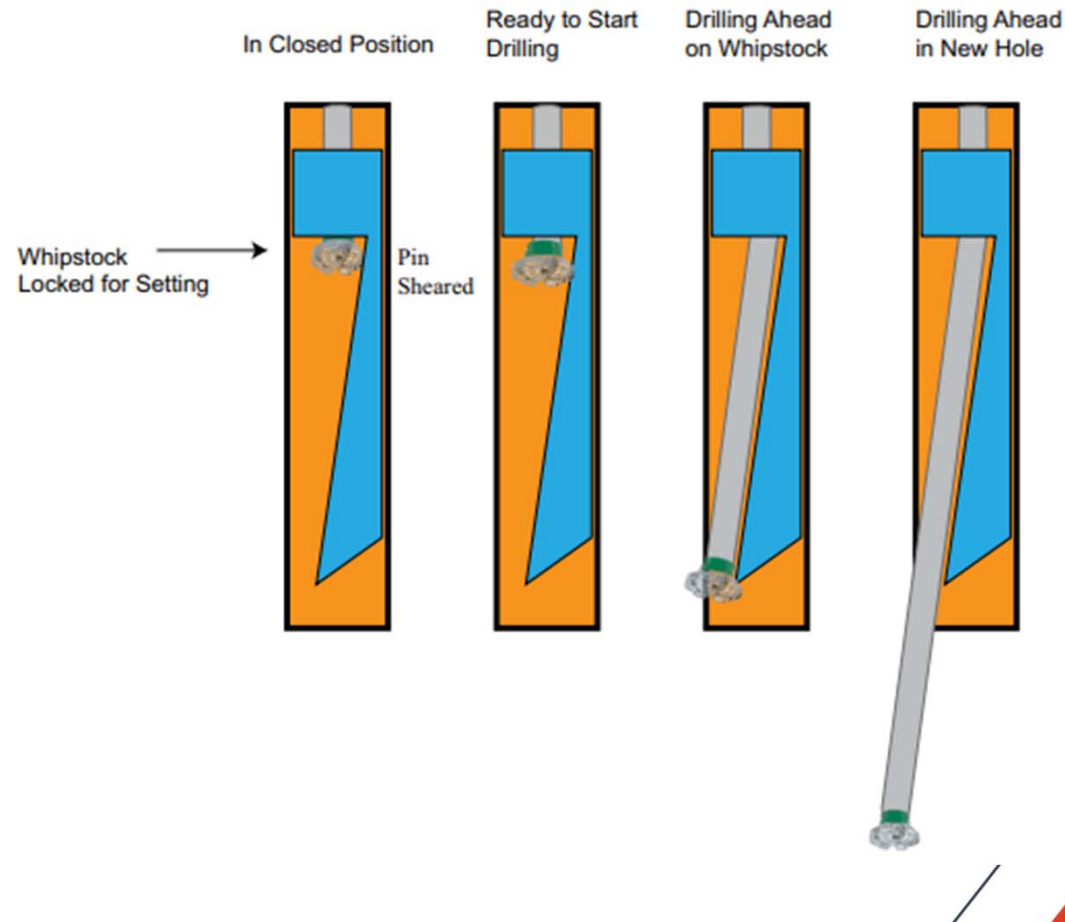
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Deflection Method - Jetting



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Deflection Method - Whipstock



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Deflection Method – Turbines

Down-Hole Turbines

Uses:

- Improve ROP
- Enable directional work

Benefits:

- No elastomers - good downhole life
- No temperature limitations



Limitations:

- High shaft speeds (rpm) require high pressures.
- Not as versatile as the PDM

Surveying Tools

How to Determine Position

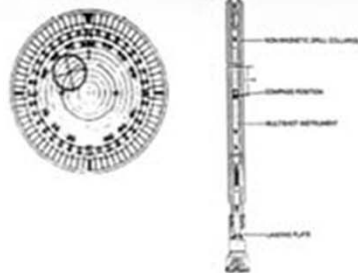
Surveys

- 'Continuous'
- Intermittent
- Magnetic
- Gyroscopic

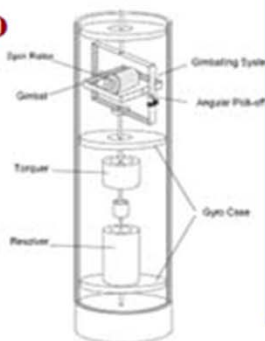
Single Shot



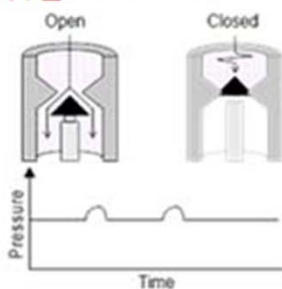
Multi-Shot



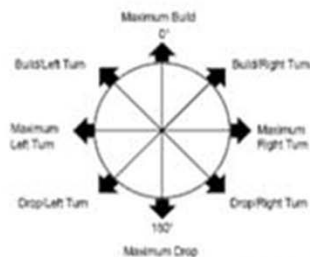
Gyro



MWD Positive Pulse

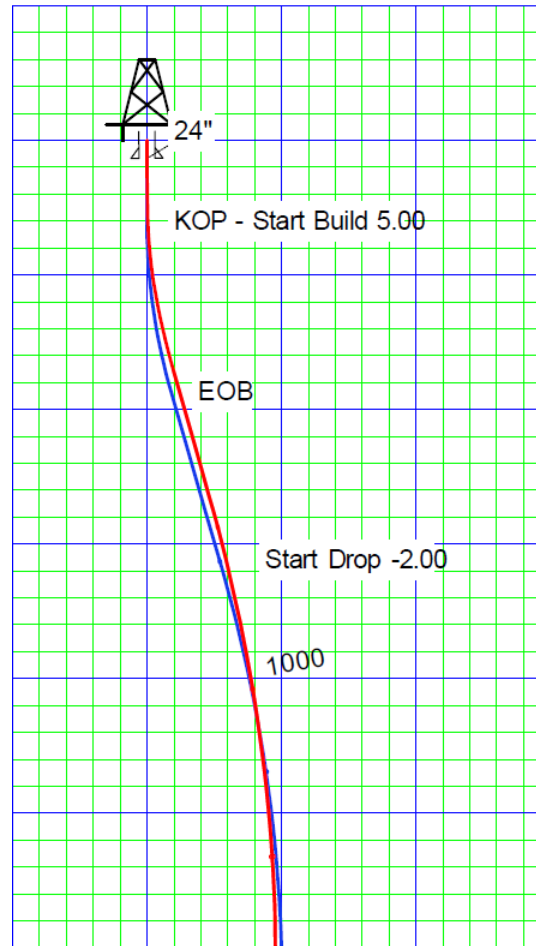
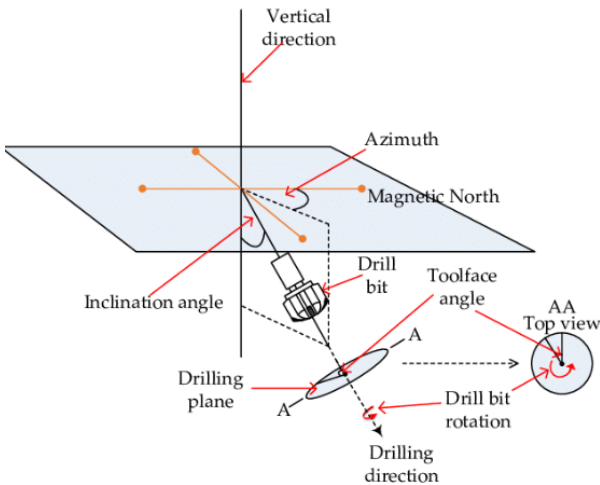
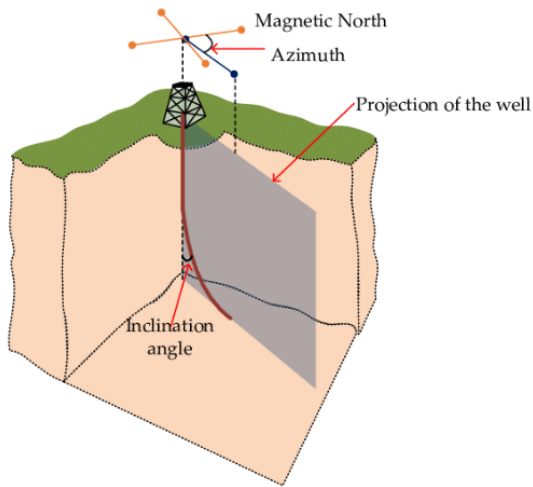


Tool Face (High Side)



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Measurement While Drilling



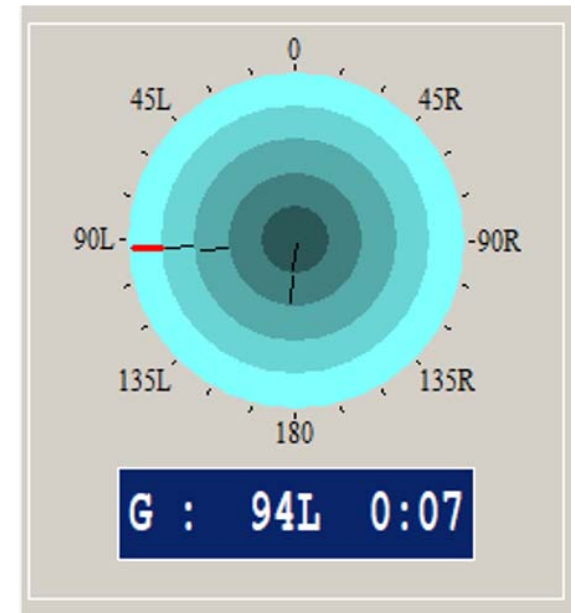
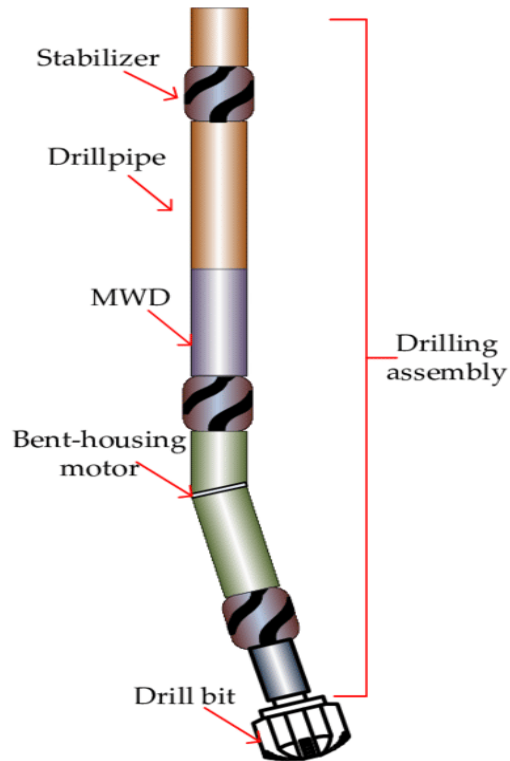
MWD

- Inclination → G's
- Azimuth → B's
- Reads Earth's magnetic field
- Affected by other magnetic source → metal, certain formation
- Short collar correction

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Benefits of MWD in BHA

- Faster directional survey and **toolface** information for directional drilling

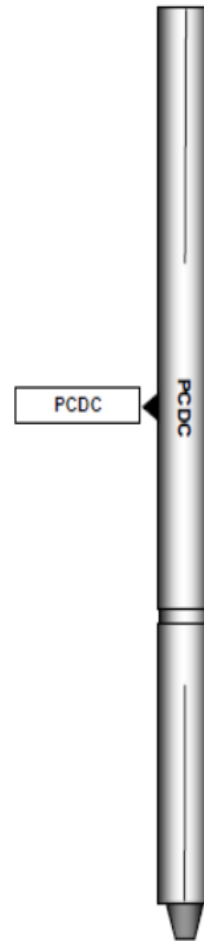


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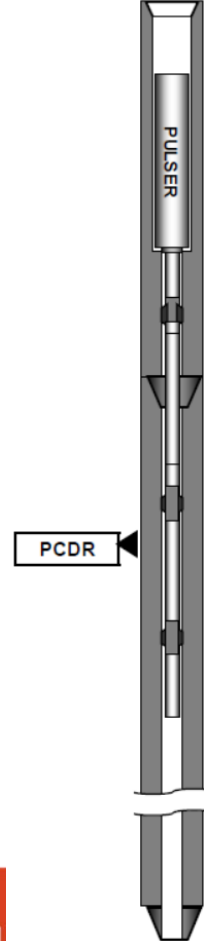
Highside Orientation Test



PM Sub



Hang of Collar

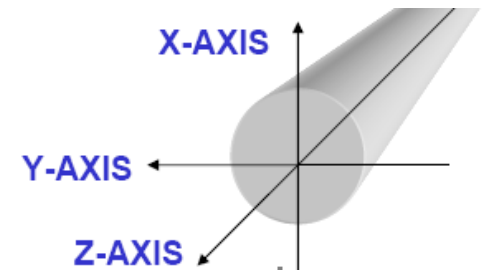


HARTON

Directional Sensor Consist of



- Magnetometers → Measure earth's magnetic field strength
 B_x, B_y, B_z
- Accelerometers → Earth's gravitational pull
 G_x, G_y, G_z

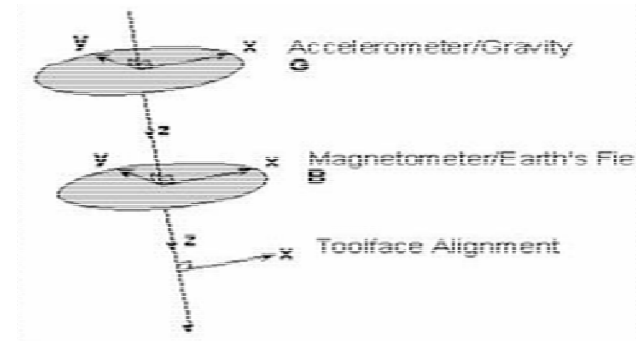


Directional Sensor



- Based on $G_x, G_y, G_z, B_x, B_y, B_z$, It can be known:

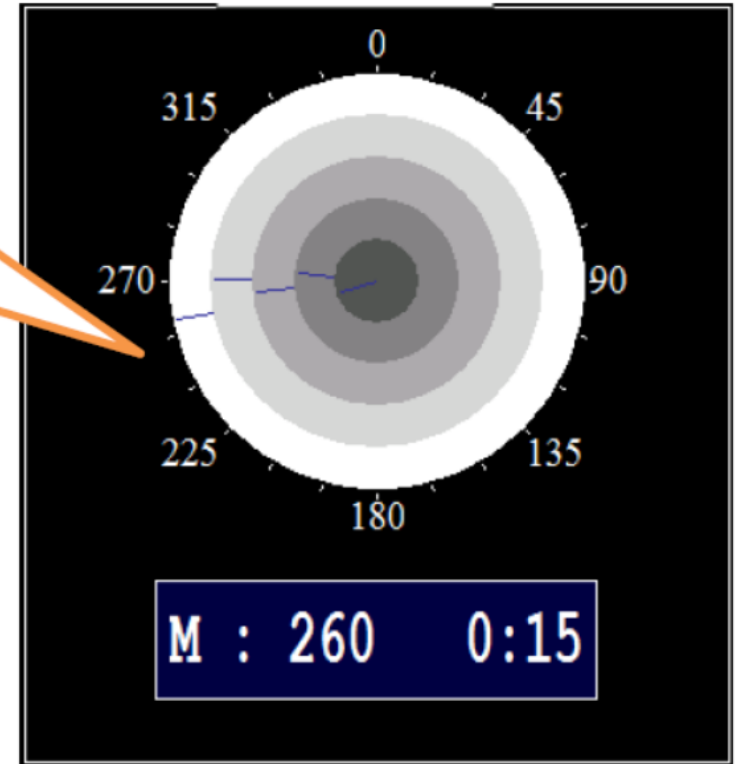
- G_x, G_y, G_z → Inclination
- $B_x, B_y, B_z, G_x, G_y, G_z$ → Azimuth
- G_x, G_y → Gravity Toolface
- $B_x, B_y, B_z, G_x, G_y, G_z$ → Magnetic Toolface



Magnetic Toolface

- Is the direction, in the horizontal plane, the bent sub scribe line is pointing with regard to the north reference. (Directional Surveying Fundamentals manual)
- Will be represented in azimuth or quadrant form, referenced to magnetic north.

Motor (Bent sub) scribe line pointing 260° from the referenced north

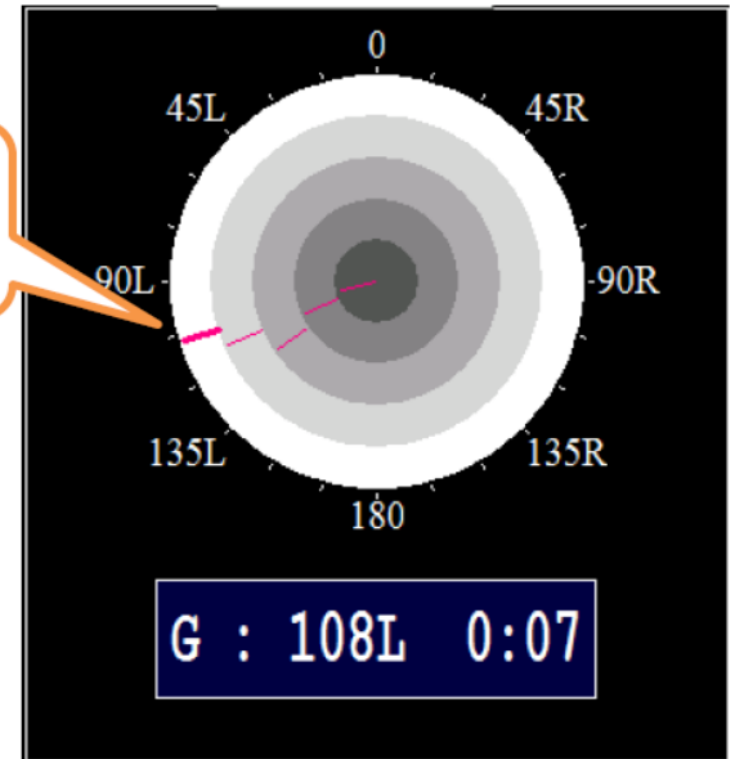


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Gravity Toolface

- Is the angular distance the motor / bent sub scribe line is turned, about the tool axis, relative to the highside of the hole. (Directional Surveying Fundamentals manual)
- Can be used if the inclination of the wellbore is above the threshold (HMS: 5 deg).
- Will be referenced to the highside of the survey instrument and presented in a number of degrees either right or left of the highside.

Motor (Bent sub) scribe line pointing 105° left from the high side of a directional tool



North Reference



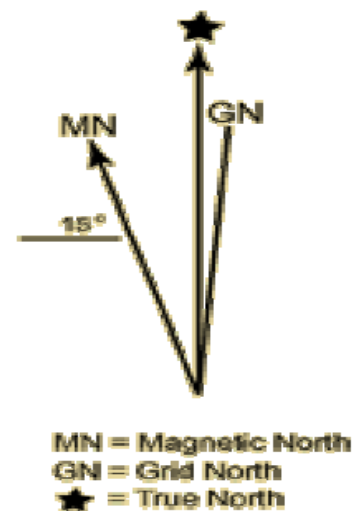
There are:

- Magnetic North (Dec)
- True North
- Grid North (Conv)

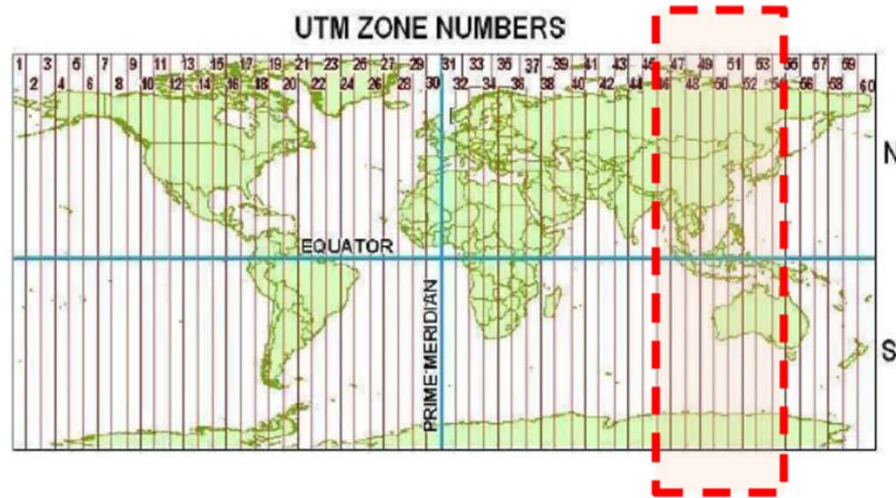
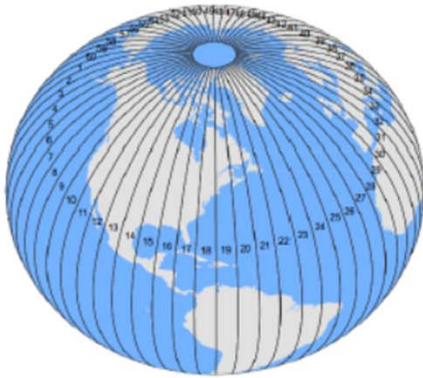
Purpose:

- Determine the total Correction (conv and dec) due to our reference
- Basic Formula:

$$\text{Grid Direction} = (\text{Magnetic Direction} + \text{Declination}) - \text{Convergence}$$



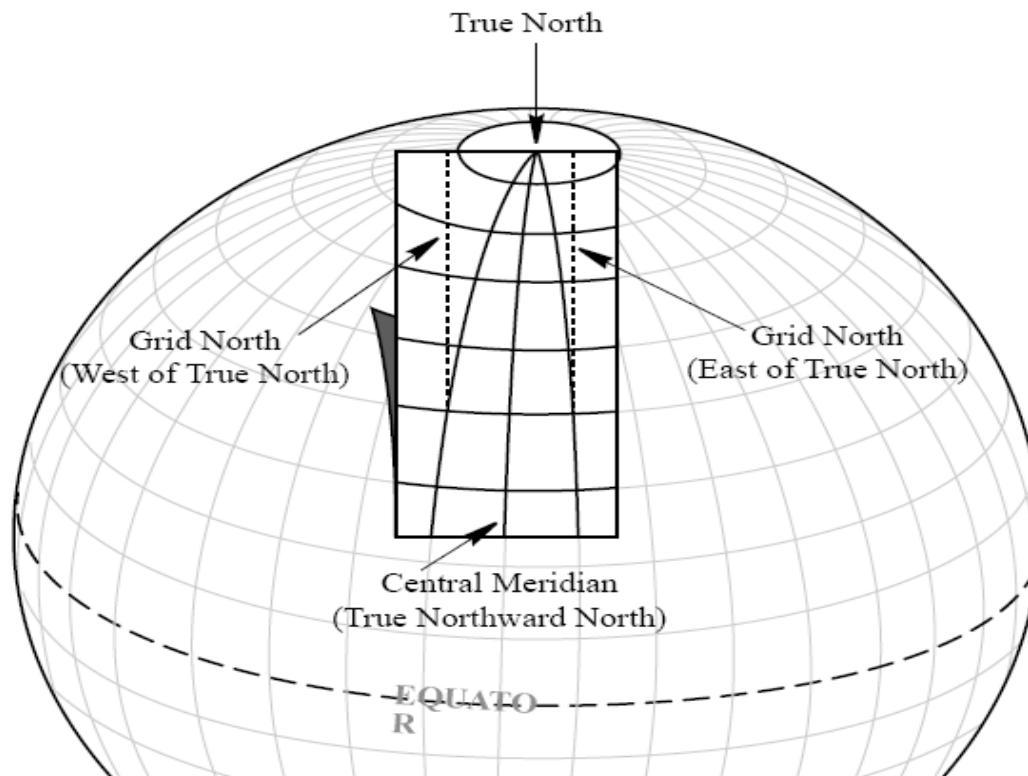
Coordinate Reference: UTM



Indonesia : Zone 46 – 54

Mercator/UTM Projection divided to be 60 zones
Each zone, 6° wide, is projected separately to grid map

MagUTM

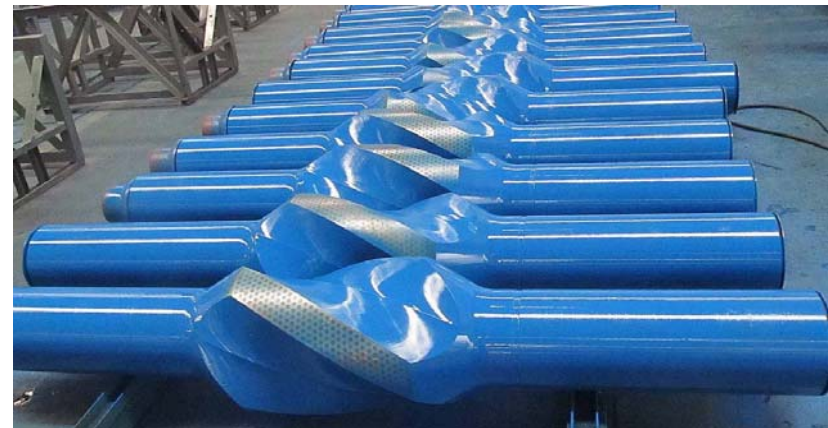


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Stabilizer

Functionality

- Controls contact point between hole and the BHA
- Used to maintain a hole angle or change angle
- **Types**
 - Sleeve stabilizers
 - Welded-blade stabilizers
 - Integral-blade stabilizers (IBS)
 - Roller reamers
 - Adjustable Gauge Stabilizer (AGS)

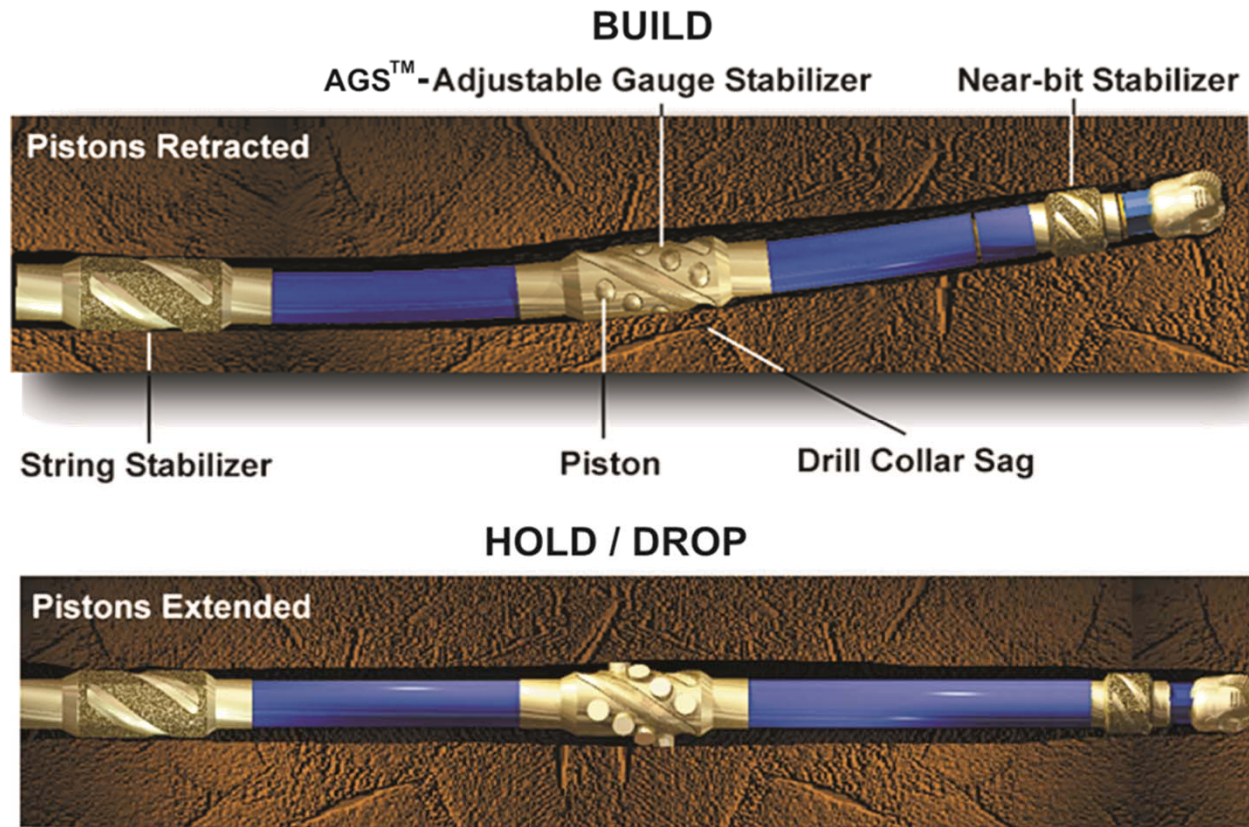


Adjustable Gauge Stabilizer

- AGS tool provides optimum directional control for rotary and steerable assemblies
- Allows changes in the build or drop tendencies without tripping to reposition stabilizers.
- By toggling between two different positions, the inclination tendency of the BHA can be manipulated while in rotary mode.



2D Control with AGS Tool

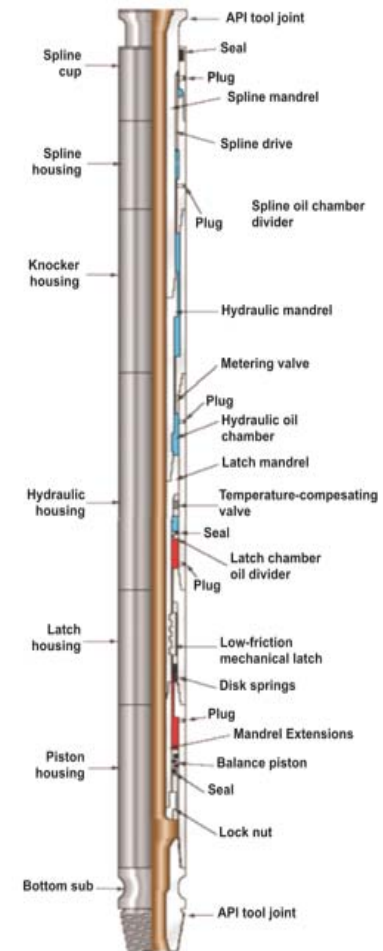


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Drilling Jar

Functionality

- Impact tool located in the drill string used to free stuck pipe
- Most effective when the drill string is mechanically restrained by the formation and not differentially stuck due to mud imbalance
- Types of drilling jars; each refers to trigger mechanisms available
 - Hydraulic
 - Mechanical
 - HydroMechanical



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Drilling Jar Intensifier

- Jarring accelerators are used to increase the chance on recovery of a stuck-pipe incident.
- Jarring accelerator stores strain energy over a significantly less length than that required for the drillstring. This is accomplished by compressing a series of springs inside the tool or compressing gas or a compressible fluid, such as silicon, inside the tool.
- **Can we run jar without accelerator ???**

LETS DISCUSS



Drilling BHA – Drill Pipe



- Connects the BHA to surface for hoisting
- Facilitates drilling and tripping
- Transmits torque from surface to BHA
- Transmits hydraulic energy from surface to the BHA



Drilling BHA – HW Drill Pipe

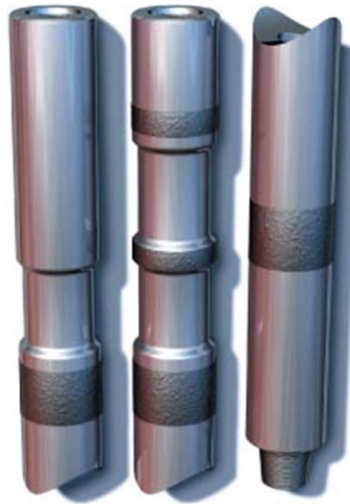


- Has the same OD of DP and reduced ID
- Used between DP and DC
- Sizes: 3½, 4, 4½, 5, 5½, 6-5/8 inch and in length of 30.5 feet
- Can be safely run in compression
- Used to provide WOB in directional and horizontal wells
- Number of joints used varies for each BHA. Normally driven by Jar Placement analysis and BHA design in general



Drilling BHA –Drill Collar

- Thick walled pipe used to provide WOB to keep DP in tension
- Two common designs, Slick and Spiral type
- Pin and Box are cut from the body of the pipe



Drilling BHA –NM Drill Collar

- Function the same as regular Drill collar
- Non-Magnetic material is used to provide spacing for Magnetic Survey tool (MWD)



Subs



XO Sub

Circulating Sub

Dump Sub
(old school)

Bit Sub (FS)

Float Sub

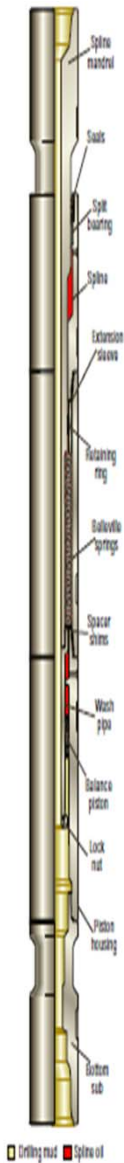
UBHO



Shock Tool / Shock Sub



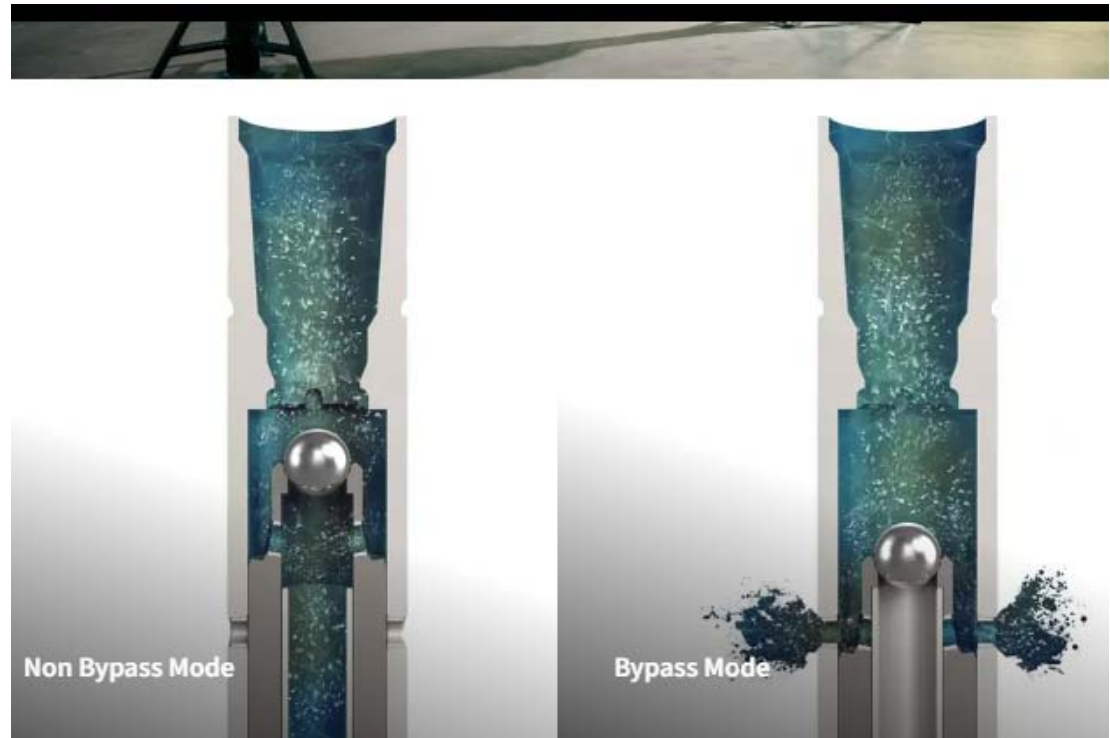
- Used to combat axial bit vibration in order to improve run length and penetration rate.
- Should only be used when:
 - **Drilling hard formations**
 - In combination with tri-cone bits
 - **In low angle sections (<30deg)**
- Only effective when placed as close to the bit as possible.
- Two key parameters, which determine the effectiveness of the shock sub, are its **spring rate** and **stroke length**.



2 types :
Mechanical shock sub
Hydraulic shock sub

Special / Performance Tools – Circ.Sub

- Drop ball activated
- Some design have limited cycles, unlimited cycles also available
- Latest technology use downlink to open and close by-pass valve
- (<https://mit-technologies.com/icwd/>)



Special / Performance Tools – Friction Reduction Tools

- Reduce Friction along the drillstring
- Different provider have different methods (axial and lateral oscillation)
- Some provider use Power Section like Rotor-Stator assy to assist generation of oscillation force
- Also can be utilized in Fishing Operation



Hole Opening and Under-Reaming Defined

- Both Considered “Hole Enlargement”
- **Hole Opening** is simply enlarging the well bore (hole) either as a dedicated run or simultaneous enlargement (with bit and hole opening tool in the string at the same time)
- **Under-Reaming** is enlarging the well bore (hole) below a restriction of some type either as a dedicated run or simultaneous enlargement (with bit and under-reamer aka “reamer” in the string at the same time)

Hole Enlargement Applications



- Facilitate running of multiple strings of casing
 - Short radius directional jobs (to enlarge the curve section) and allow easier and better casing runs
 - Enlarge above a RST (rotary steerable tools)
 - Enlarge the rathole from below the reamer to 3ft. from TD
 - Flowing/unstable formations (salt creep)
 - Expandable casing programs (requires larger holes than conventional casing)
 - Reduce ECD and pressure on formation
 - Mono Diameter well bores (large future potential)
- Enlarge above formation evaluation tools (MWD/LWD) or after coring runs

Hole Opener Types

- Function
 - Hole enlargement
 - Maintain hole diameter across sloughing shale's or swelling/collapsing formation
- Normally used in soft & medium-hard formations, spud assemblies
- Main types: Roller Cone and Fixed Blade



Roller Cone Hole Openers

- Pros

- Replace expensive larger roller cone bits
- Available in a wide range of size to 60"
- Able to be used with smaller rigs

- Cons

- Risk of Lost Cones
- Not typically designed for drill out capability (new generation however)

HOLE OPENERS



Fixed Bladed Hole Openers

■ Pros

- Wide range from pilot hole size to reamed size
- Available in a range to 26" and larger
- Reliable with no moving parts

■ Cons

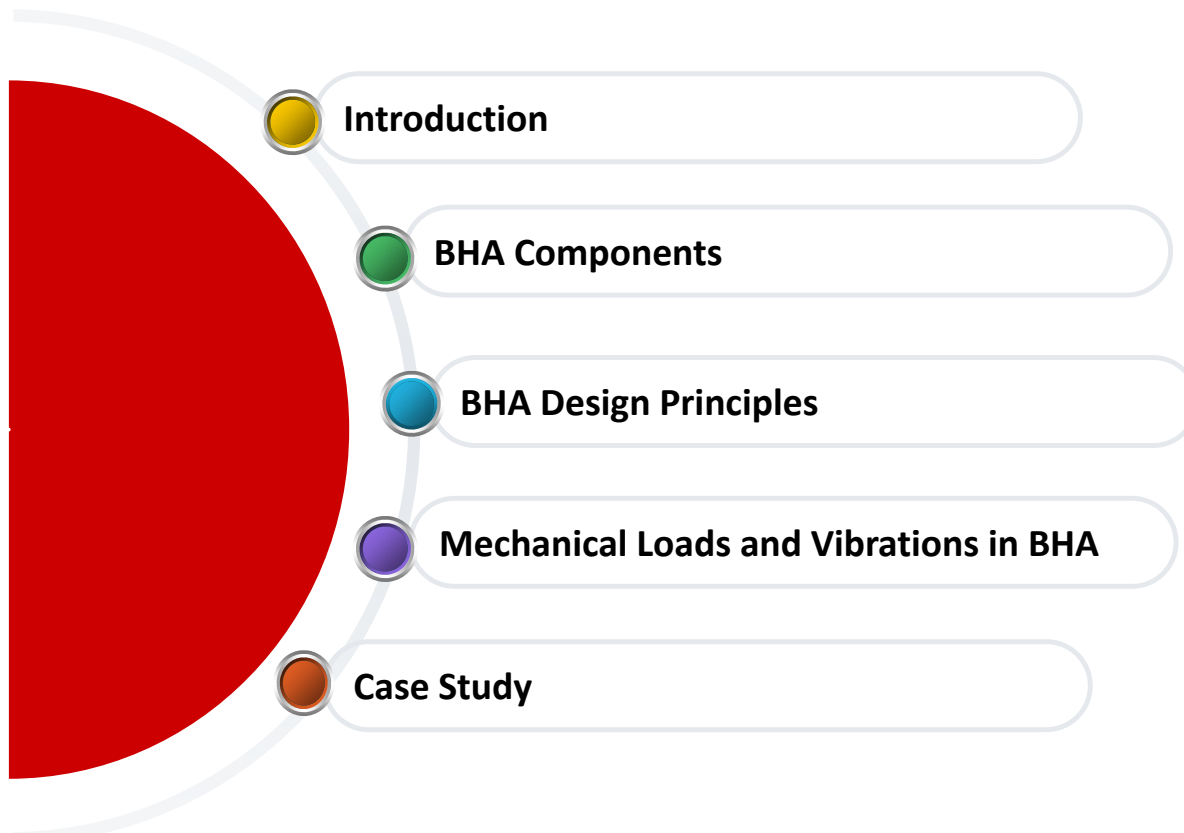
- Typically requires 2 passes (pilot hole then enlarge)
- Not typically designed for drill out capability (new generation tools may be used in that fashion however)



Special / Performance Tools – Under Reamer

- There are few activation methods, but the most common is drop ball activation. RFID option also available

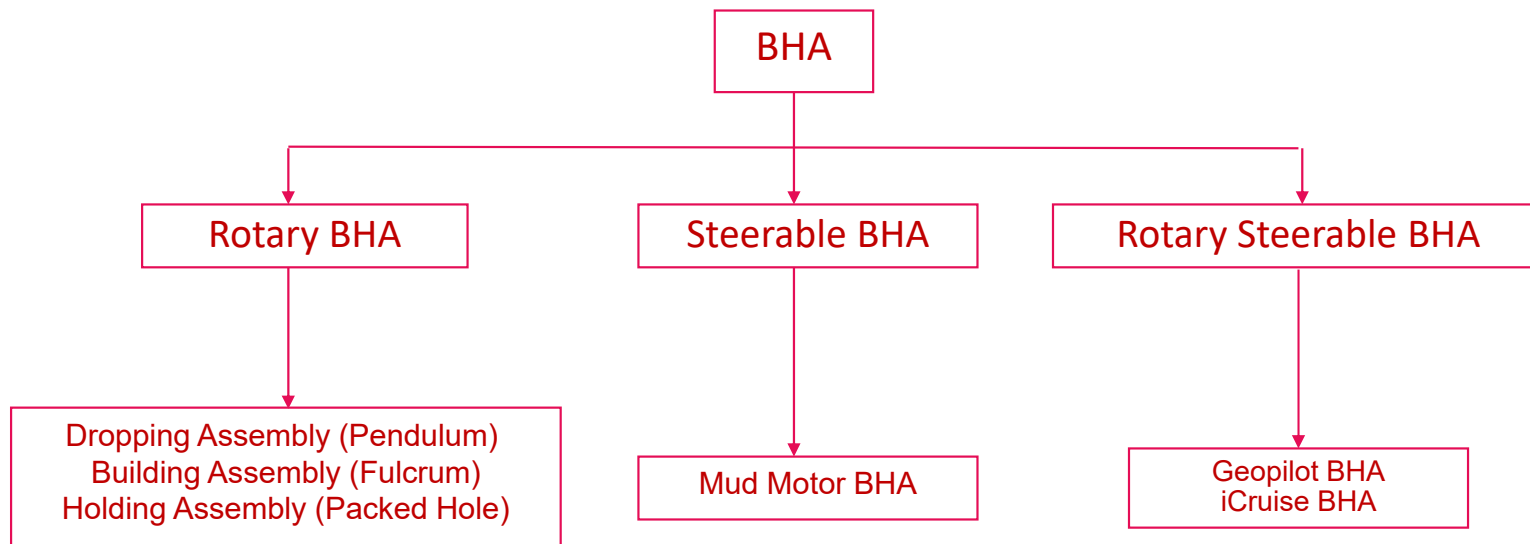




Steps in BHA Design Workflow

- 1 Obtain the required data
- 2 Decide the objectives of the BHA: Hold? Build? Drop?
- 3 Select the bit drive system
- 4 Select the bit
- 5 M/LWD magnetic spacing and survey instrument
- 6 Jar placement recommendation
- 7 Hole gauge and stabilizer gauge effect in BHA behavior
- 8 Calculate length of drill collar required for maximum WOB

Basics of Directional Drilling BHA

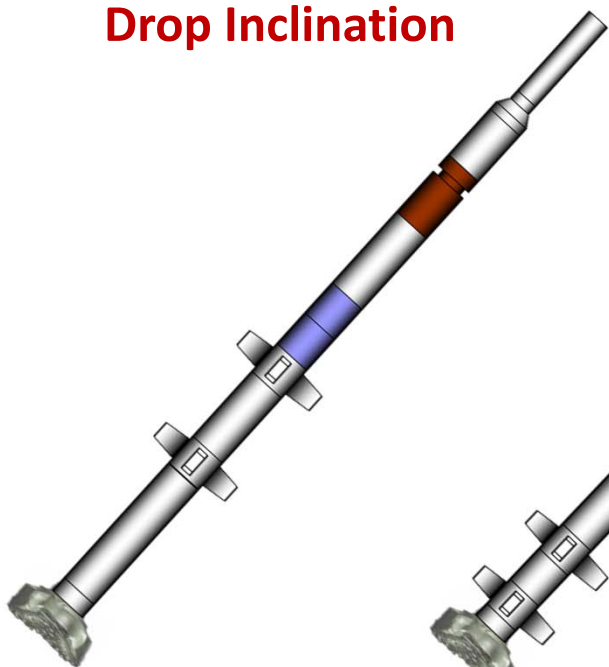


Rotary BHA



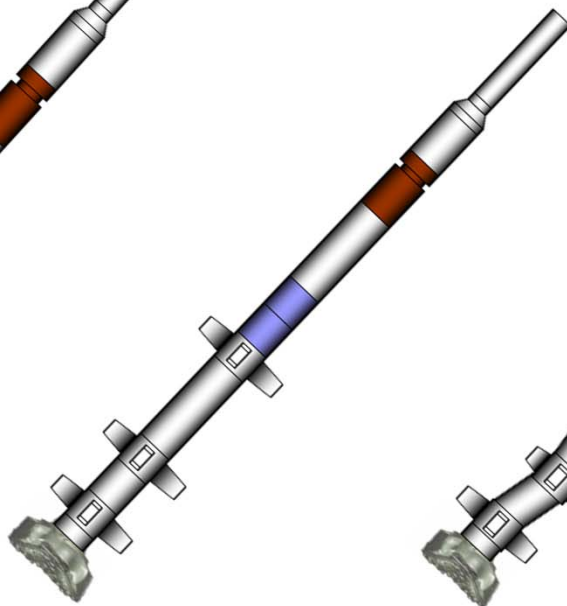
PENDULUM

Drop Inclination



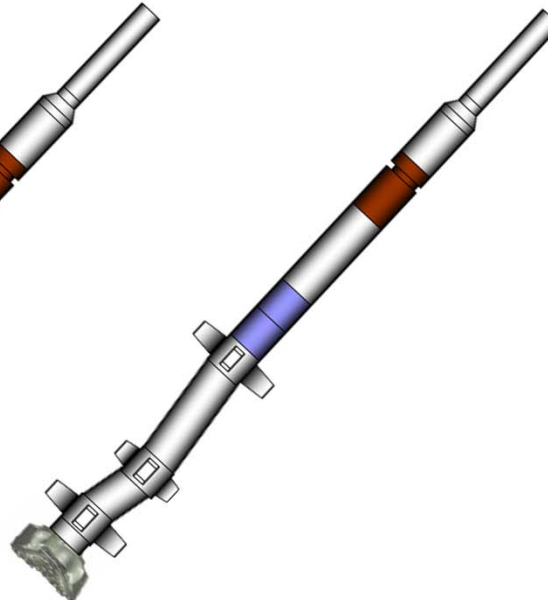
PACK

Hold Inclination



BUILD

Inclination



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Rotary BHA

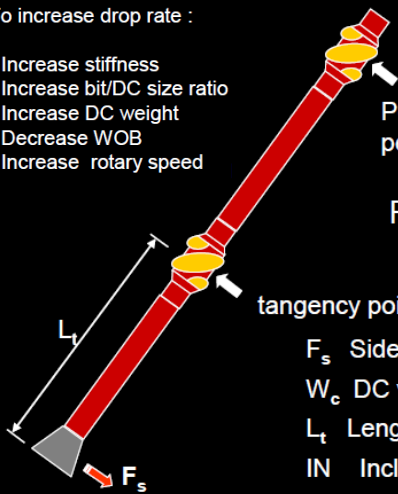


Pendulum BHA

The Pendulum Principle

To increase drop rate :

- Increase stiffness
- Increase bit/DC size ratio
- Increase DC weight
- Decrease WOB
- Increase rotary speed



Dropping Assembly

Pendulum assemblies drop angle as a pendulum creates negative side force

$$F_s = \frac{1}{2} W_c \times BF \times L_t \times \sin IN$$

F_s Side force on bit (lbs)

W_c DC weight below tangency point (lbs/ft)

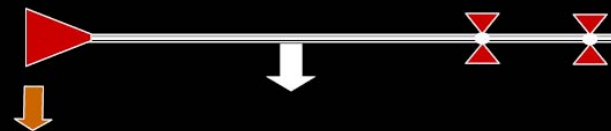
L_t Length of pendulum (ft)

IN Inclination at tangency point (deg)

BF Buoyancy factor (-)

Generic Angle-Dropping (Pendulum) Assembly

- Bit
- Drill collar, 30-90ft
- String stabilizer, full gauge
- Drill collar, 30ft
- String stabilizer, full gauge
- Drill collar, 90ft
- HWDP, 90ft min.
- Drill pipe



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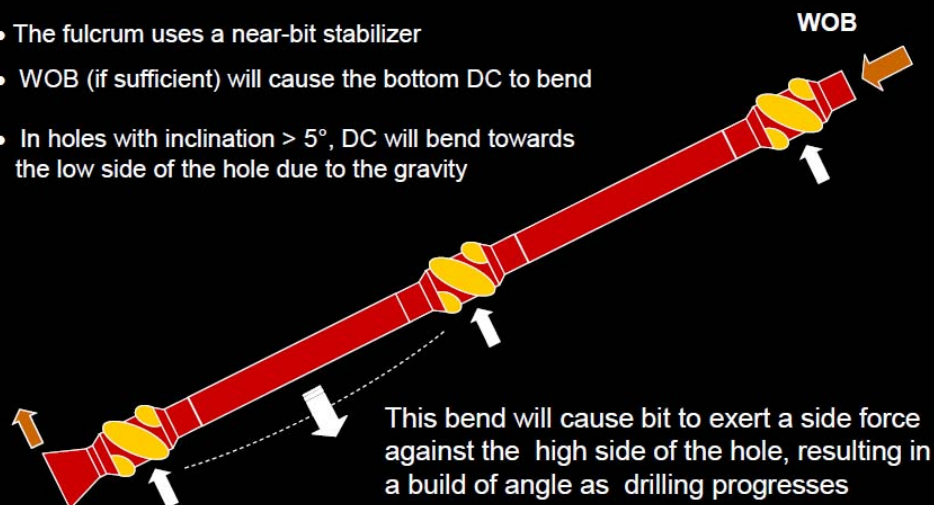
Rotary BHA



Fulcrum BHA

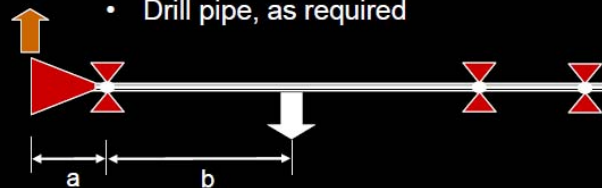
The Fulcrum Principle

- The fulcrum uses a near-bit stabilizer
- WOB (if sufficient) will cause the bottom DC to bend
- In holes with inclination $> 5^\circ$, DC will bend towards the low side of the hole due to the gravity



Generic Angle-Building (Fulcrum) Assembly

- Bit
- Near-bit Stabilizer, full gauge
- Drill collar, 30-90ft
- String stabilizer, full gauge
- Drill collar, 30-60ft
- String stabilizer, full gauge
- Drill collar, 90ft
- HWDP, 90ft min.
- Drill pipe, as required



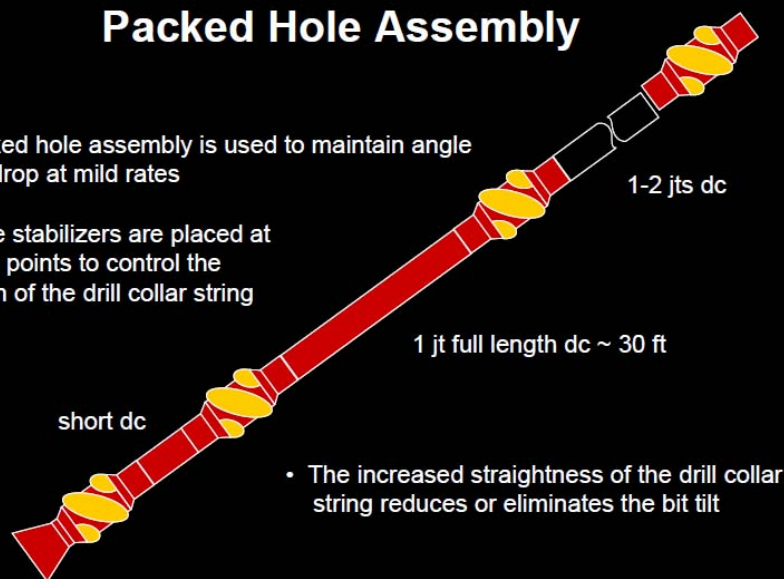
Rotary BHA



Packed BHA

Packed Hole Assembly

- The packed hole assembly is used to maintain angle or build/drop at mild rates
- 3 or more stabilizers are placed at specified points to control the deflection of the drill collar string

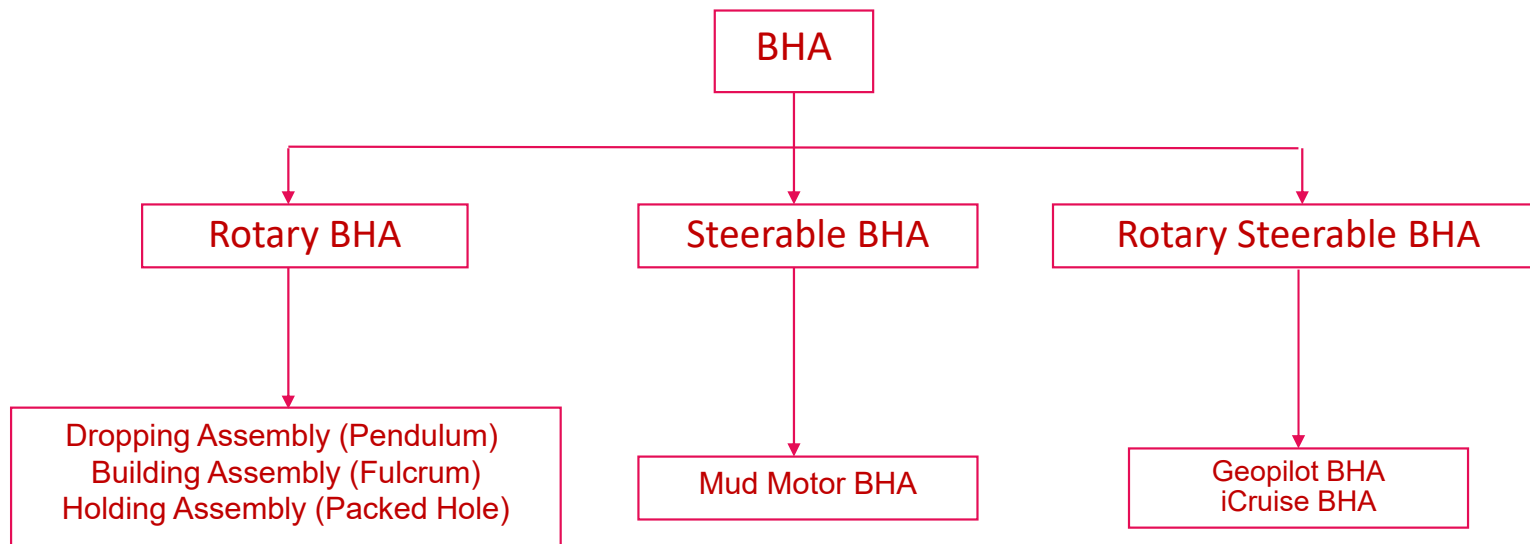


Generic Packed Hole Assembly

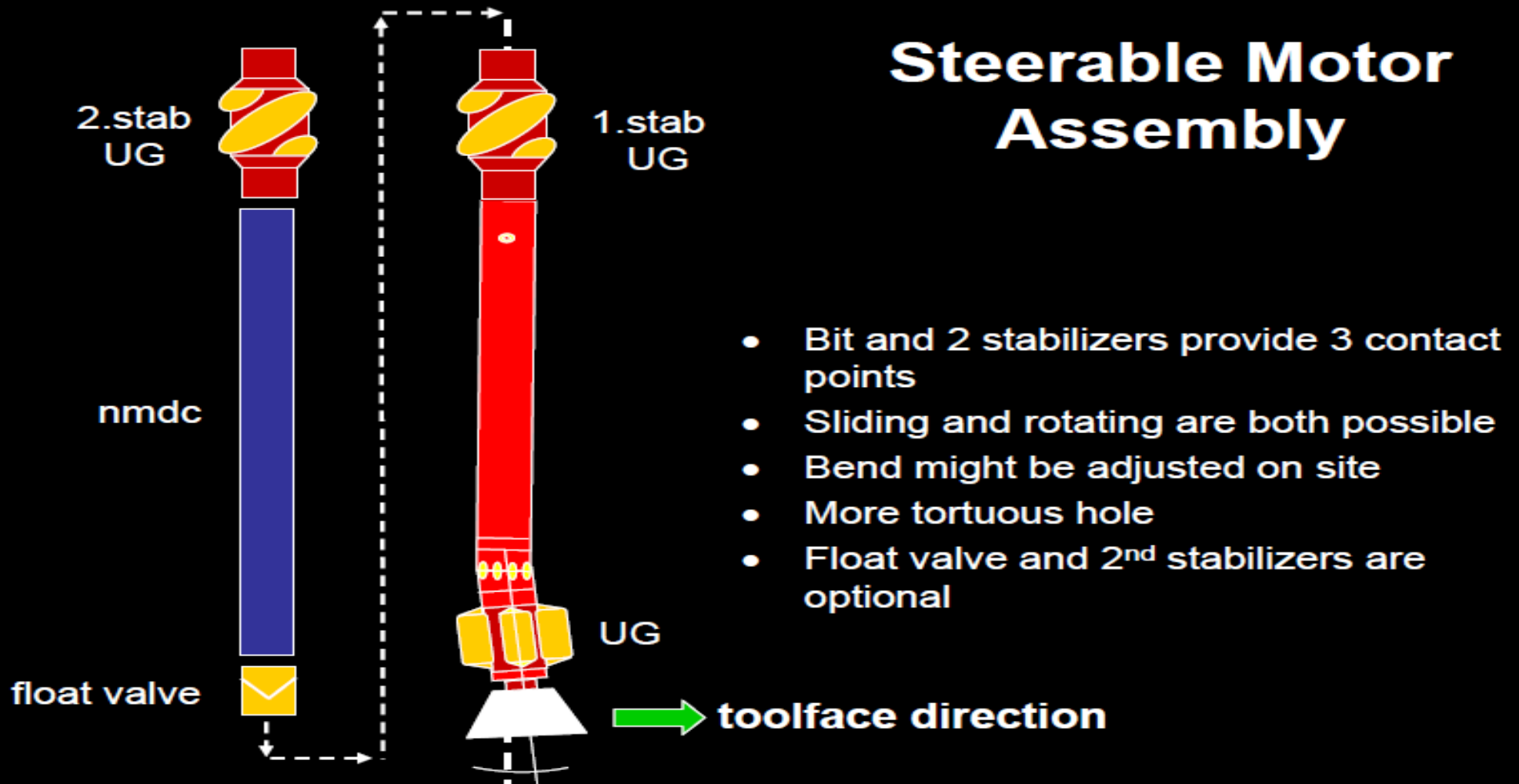
- Bit
- Near-bit Stabilizer, full gauge
- Short Drill collar, 6-20ft
- String stabilizer, full gauge
- Drill collar, 30ft
- String stabilizer, full gauge
- Drill collar, 90ft
- HWDP, 90ft min.
- Drill pipe, as required



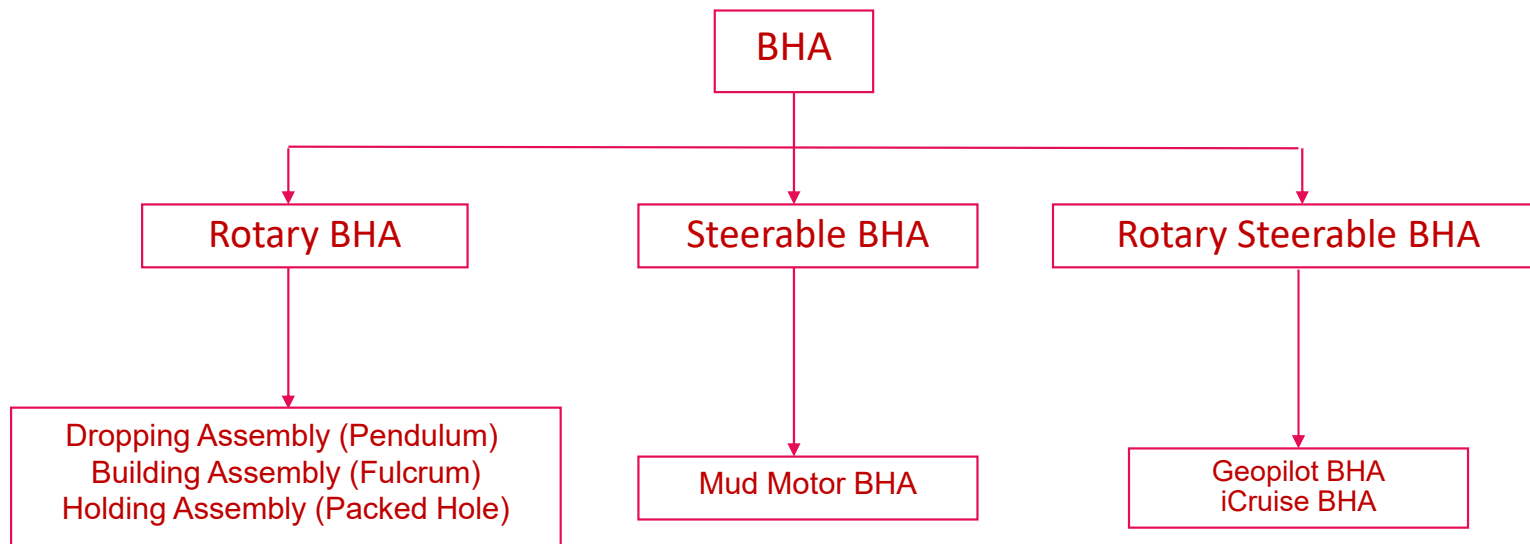
Basics of Directional Drilling BHA



Steerable BHA (Mud Motor)



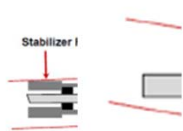
Basics of Directional Drilling BHA



Point the Bit vs Push the Bit

Point System

- Steers by precisely pointing (tilting) the bit in exactly the direction the well path needs to be steered.
- The face of the drill bit is perfectly pointing in the direction to be drilled and there is zero side loading on the bit.

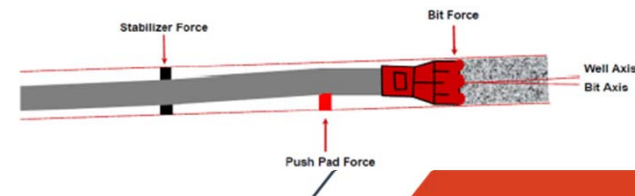
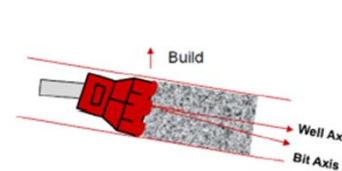


It Axis
W Axis

Push system

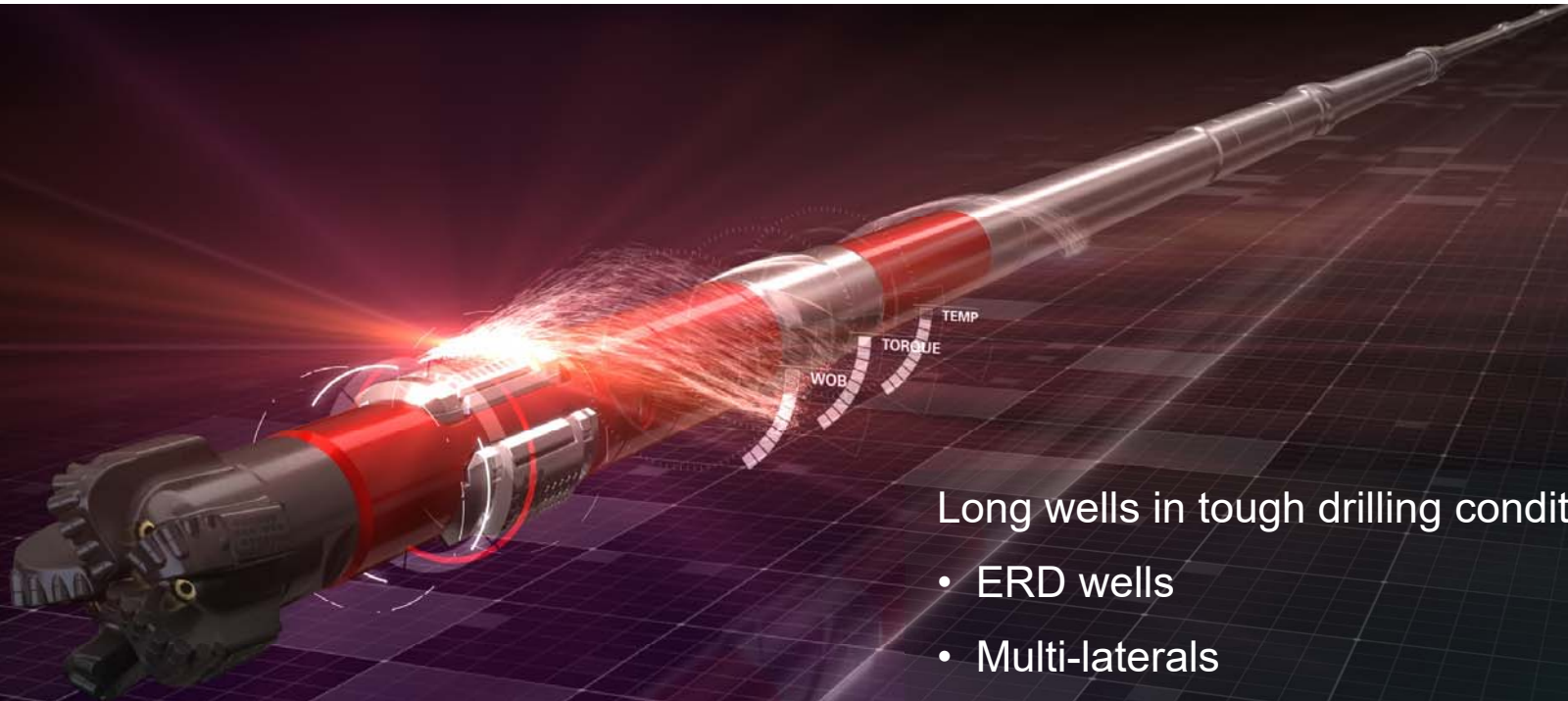
- Steers by simply applying a side load to the bit, forcing cutting structure on the outer rows and gauge of the bit to cut sideways into the formation
- Must use very short gauge bits (typically less than 2" gauge length) where the gauge is set with an active highly cutting structure.

Negative bit angle of attack (Push the bit system)



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LONGER Runs



Long wells in tough drilling conditions

- ERD wells
- Multi-laterals
- Challenging formation

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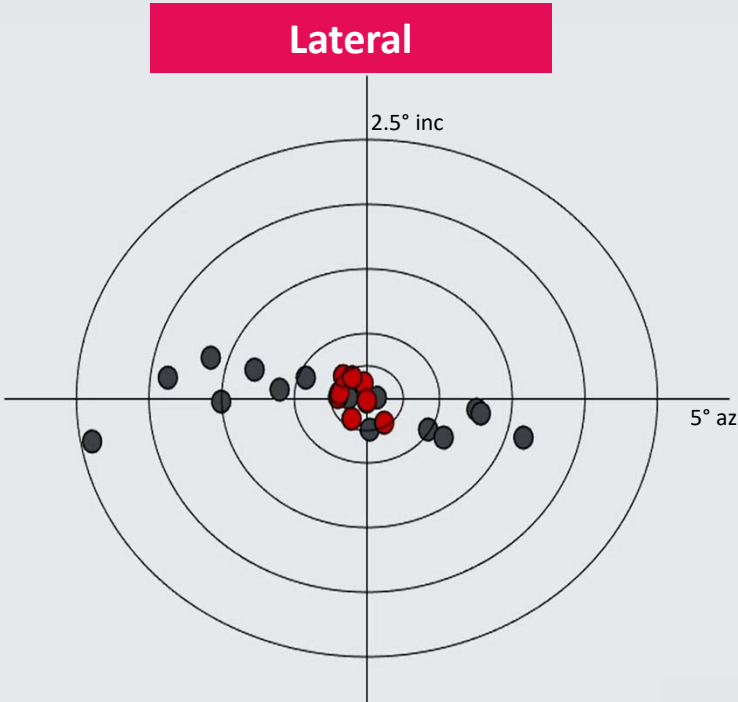
Rotary Steerable System

- Full directional control with continuous rotation of the drillstring
- High degree of steerability
- RSS optimizes
 - Increased ROP (Average)
 - Improved hole quality
 - More accurate wellbore placement
 - Improved LWD formation evaluation
- Extended
 - Reach drilling
 - Vertical drilling
 - Geo-steering
 - Concentric hole enlargement



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CruiseControl™ Technology

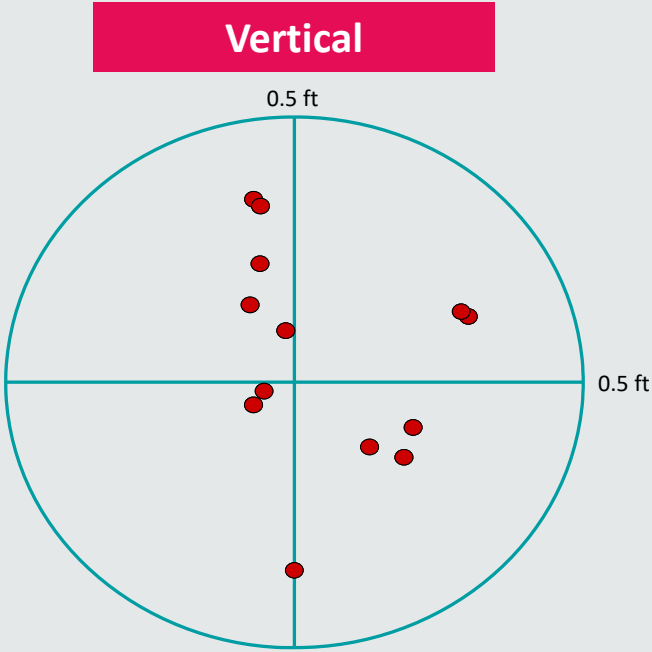


Utica Basin

- 8 ½" hole size
- 850 feet drilled with CruiseControl™
- Survey interpolated DLS <1 deg/100 ft.

Tool Center Marker

- CruiseControl™
- Manual



Norway

- 12 ¼" hole size
- 1,089 feet drilled
- Average inclination 0.1 degrees

Motors vs. RSS

• Compare

- Both are deflecting tools

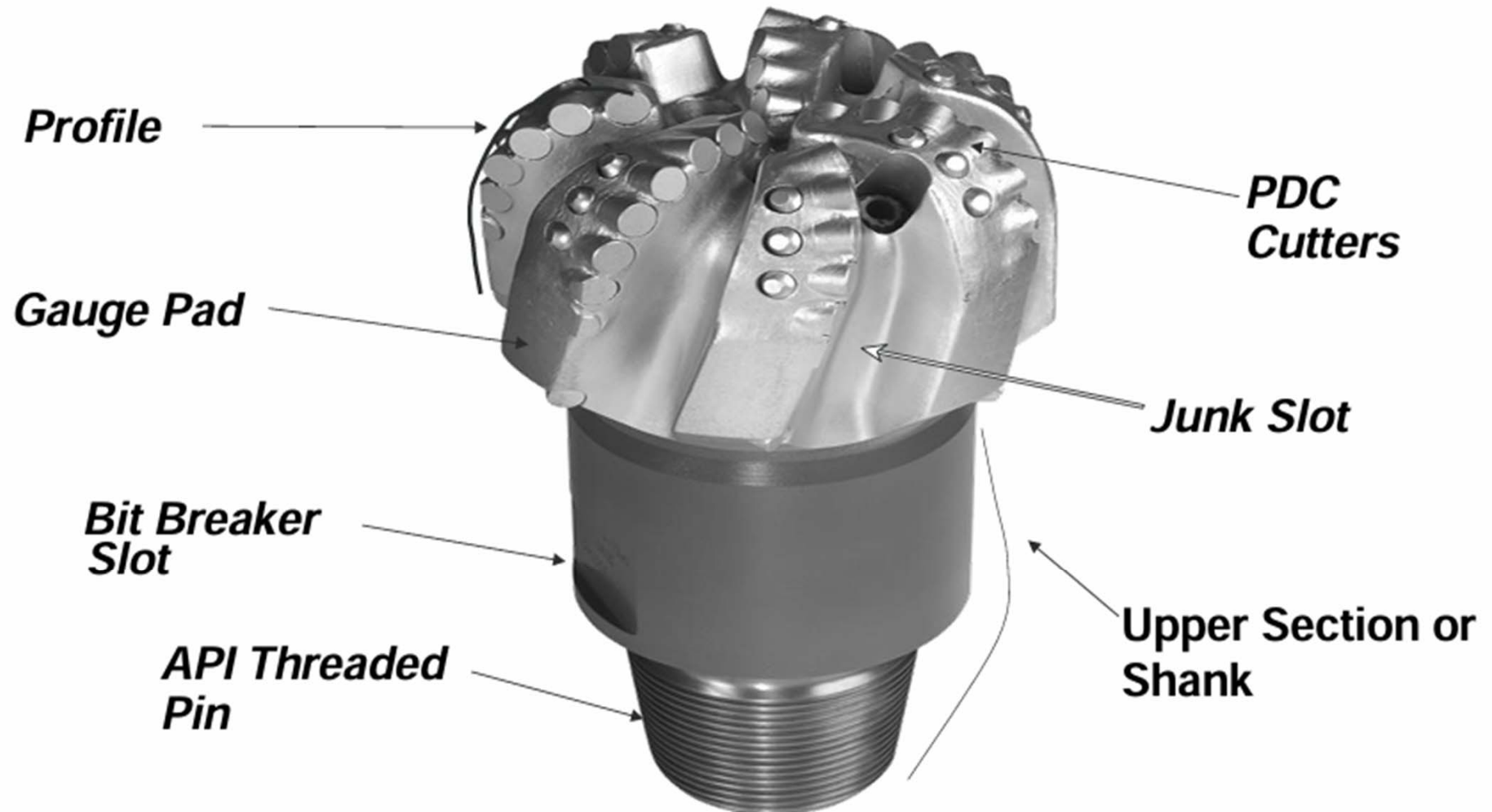
• Contrast

- Motor is a mechanical tool while RSS is a electromechanical tool
- RSS tools use a downlink to deflect
- RSS needs to communicate with the MWD tools while a motor does not
- RSS has more stringent limitations on build rates than motor
- For shorter radius curves motors are preferable to RSS
- RSS operates by rotating the drill string, hence reducing torque and drag conditions and improves hole cleaning
- Motors are better in high tortuosity conditions
- RSS is more cost effective

Bit Type (Roller Cone – Fixed Cutter)



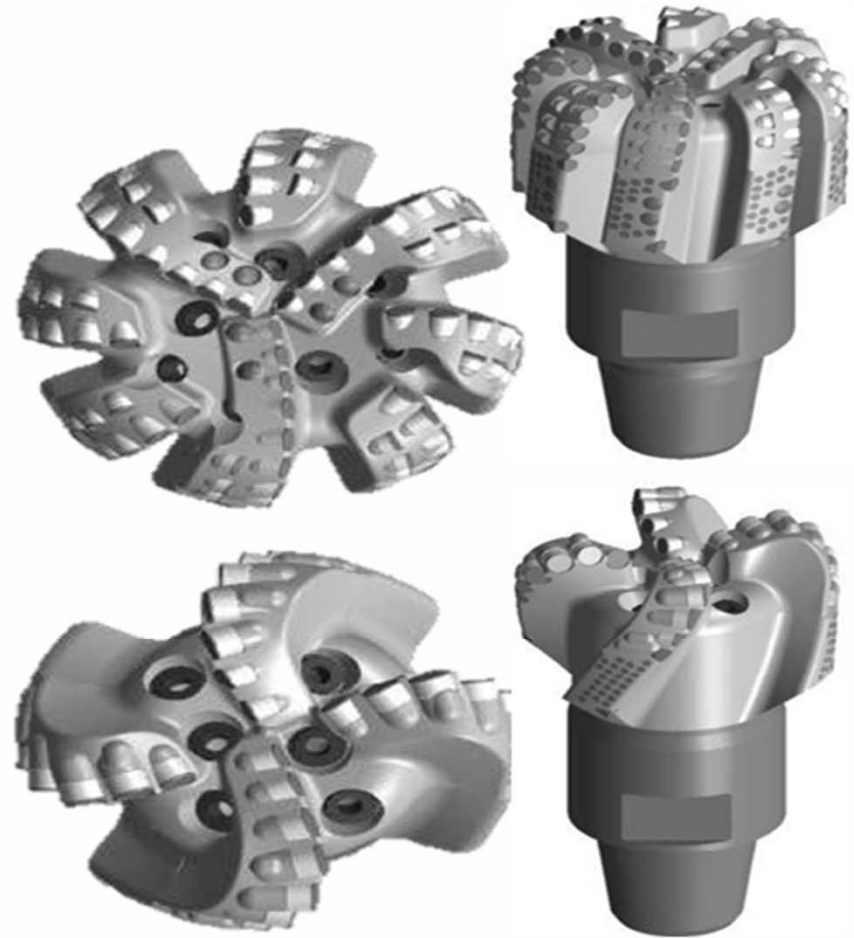
Fixed Cutter Bit Terminology



Blade Count

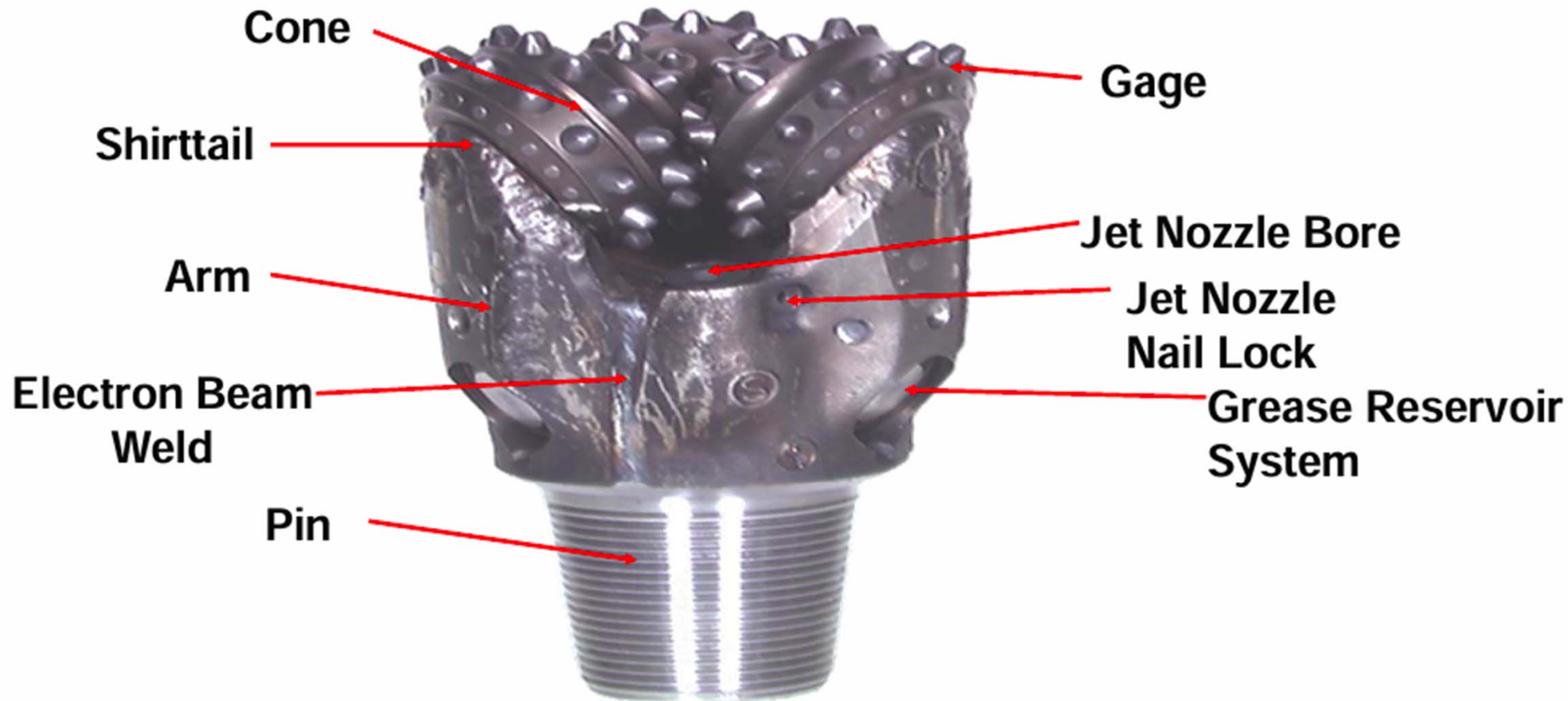
- Higher Blade Count
 - Less Aggressive
 - More Stable
 - Harder to Clean at High ROP
 - Less Nozzles

- Lower Blade Count
 - More Aggressive
 - Less Stable
 - Easier to Clean at High ROP
 - More Nozzles



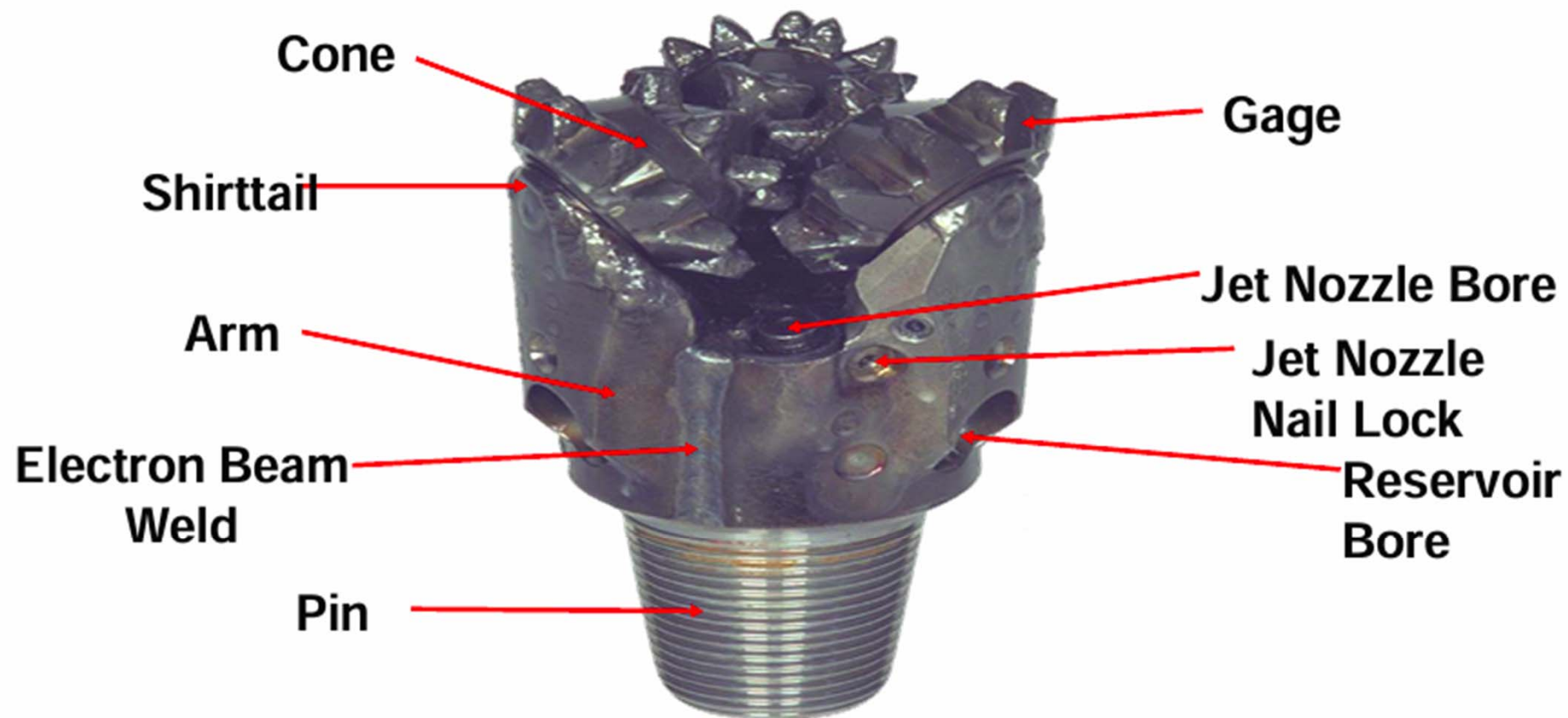
Roller Cone Bit - Terminology

Standard Frame - Insert



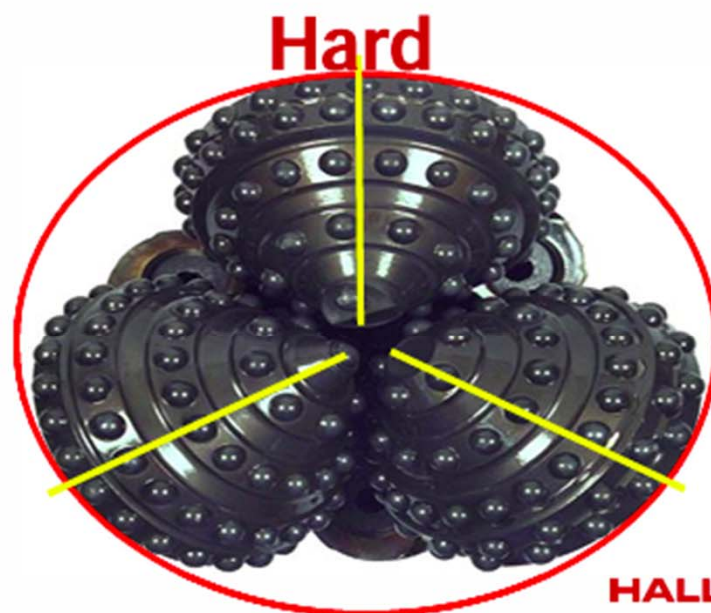
Roller Cone Bit - Terminology

Standard Frame - Steel Tooth



Roller Cone Geometry - Offset

- High offset provides increased ROP in soft formation
 - More offset increases “gouging and scraping” action on bottom)
- Less offset for medium formations
- Little or no cone offset for hard formation
 - All Weight applied for “crushing” rock



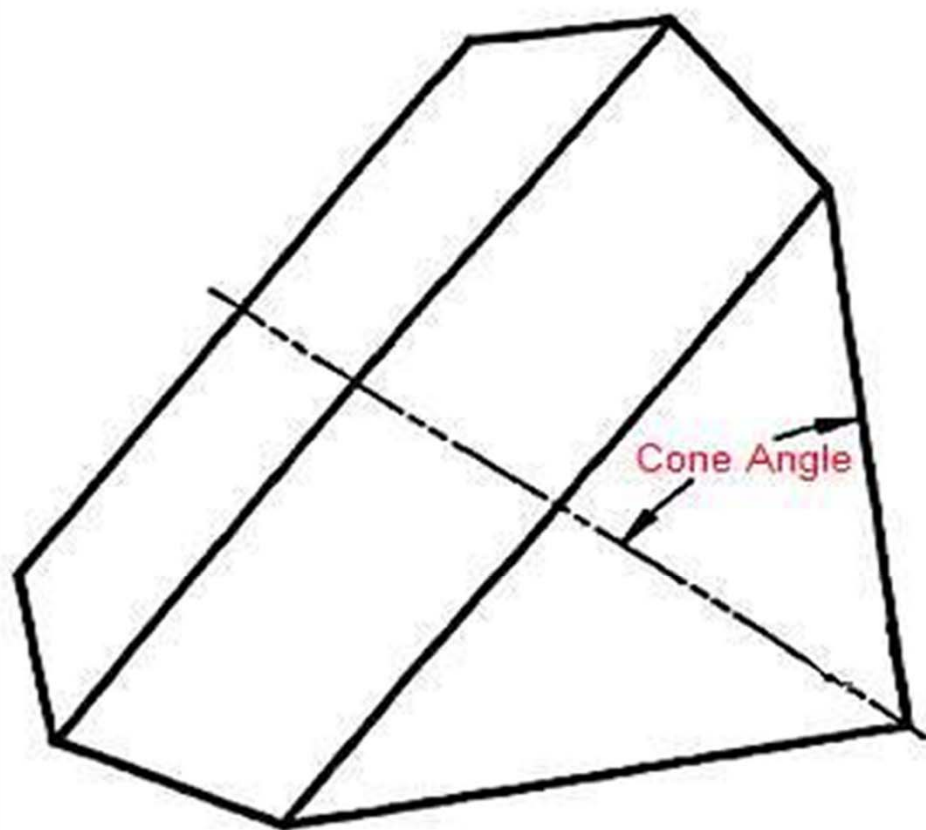
Roller Cone Geometry - Cone Angle

- Soft – Medium Formations

- Larger cone angles
- Rounder cone profile
- Gouging & Scraping

- Hard Formations

- Smaller cone angles
- Flatter cone profile
- Crushing



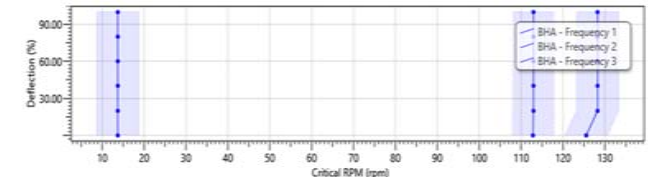
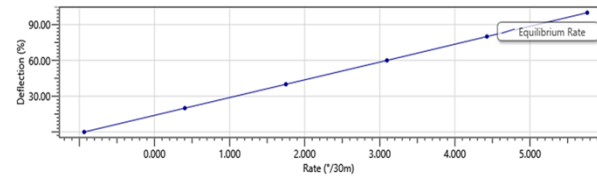
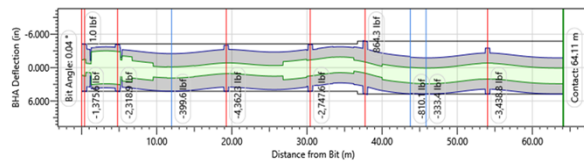
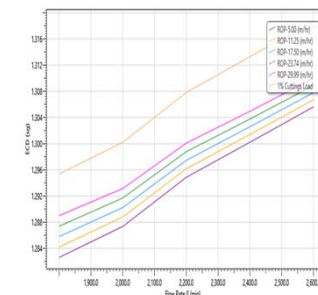
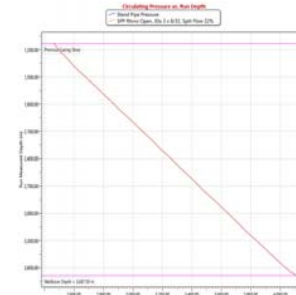
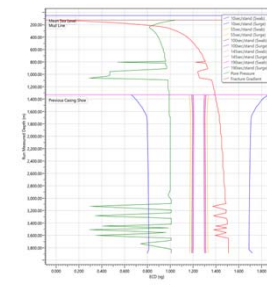
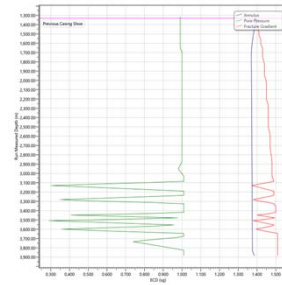
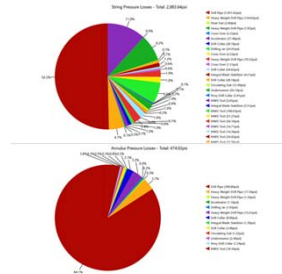
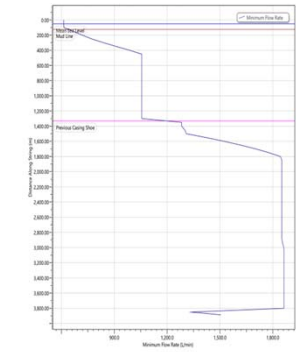
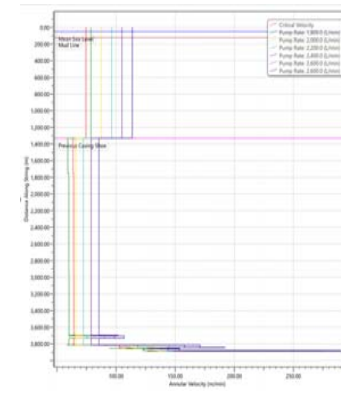
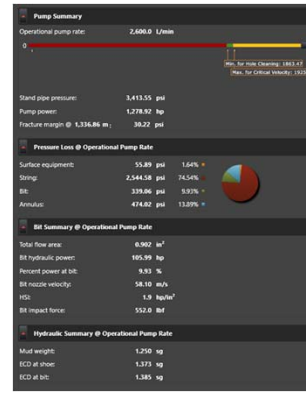
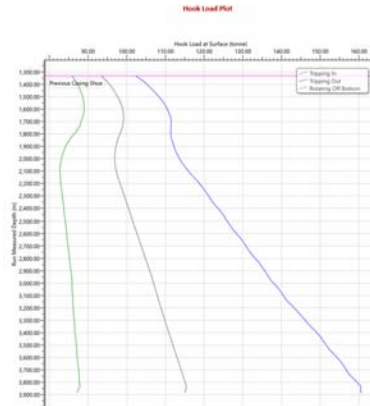
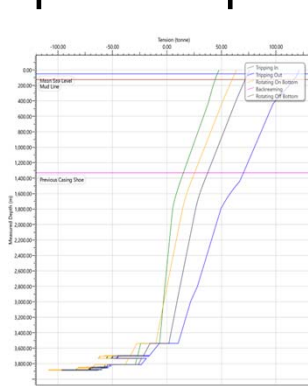
MWD Non-Magnetic Spacing

- The drillstring is comprised of ferrous material that can become magnetized and effect the magnetometer responses in the directional module.
- There are two approaches to solving this problem:
 - The first is to reduce the influence of the ferrous material by mounting the survey instrument in non-magnetic drill collar and placing non-magnetic collars above and below it. With sufficient spacing the measurements are unaffected and a “clean” reading is taken. This is known as a long collar (LC) BHA.
 - The second is to correct for the influence of the ferrous material, the survey instrument is mounted in non-magnetic drill collar but the lengths of non-magnetic collars above and below the instrument are significantly shorter. Readings of the "Z magnetometer" are affected magnetically by the drillstring but can be calculated from other responses. This is known as a Short Collar BHA and a Short Collar Correction (SCC).
- The Short Collar Correction (SCC) is a mathematical method of reducing the effects of drill string magnetic interference on magnetic survey data derived from tri-axial survey instruments.

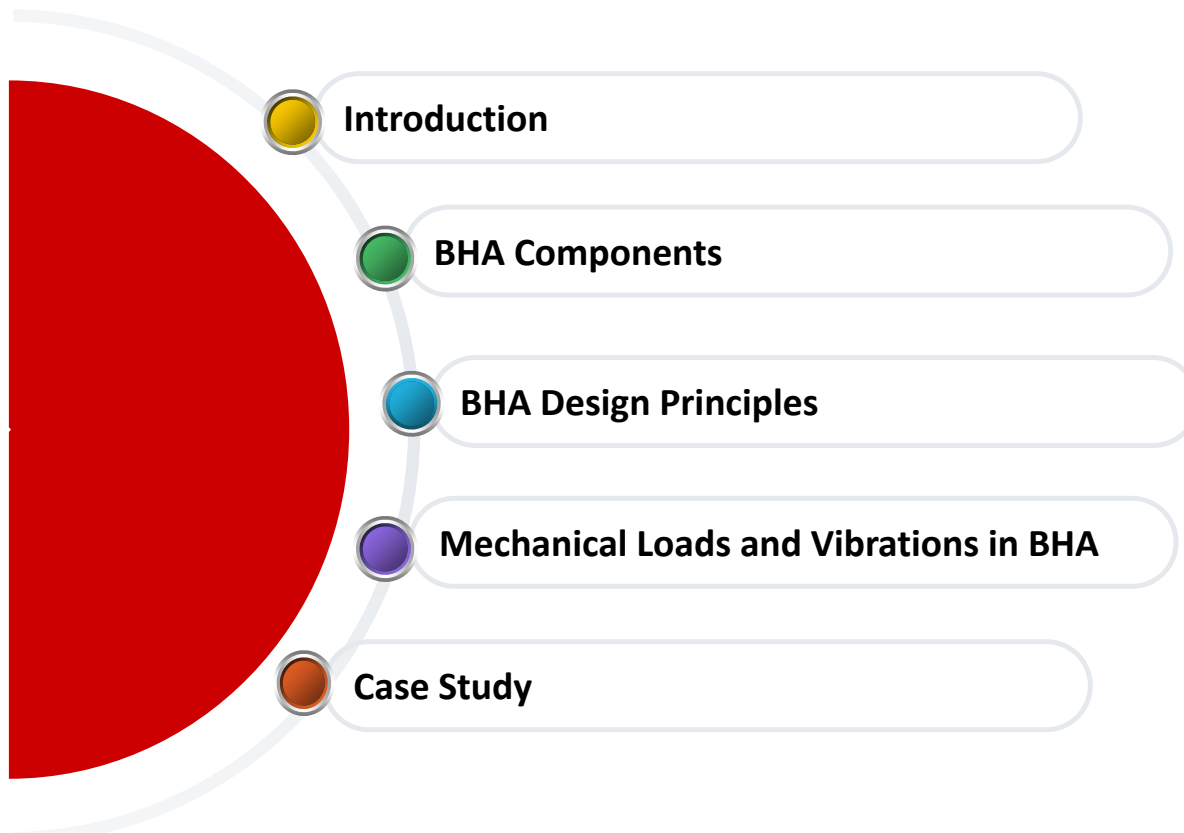
┐ BHA design – What engineering is required?

- Engineering analysis is required for each BHA design
- The verification includes:
 - BHA force modeling
 - BHA vibration modeling
 - Torque and Drag Analysis
 - Hydraulics Analysis
 - Surge & Swab

DrillingXpert Output



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Mechanical Loads and Vibration Effect



Failure Modes of the BHA



- Tensional Yield of connections
- Torsional Yield of connections
- Combined Tension and Torsional Yield of connections
- Bending separation of connection faces
- Fatigue leading to tensional or torsional failure
- Burst
- Collapse

Yield Stresses

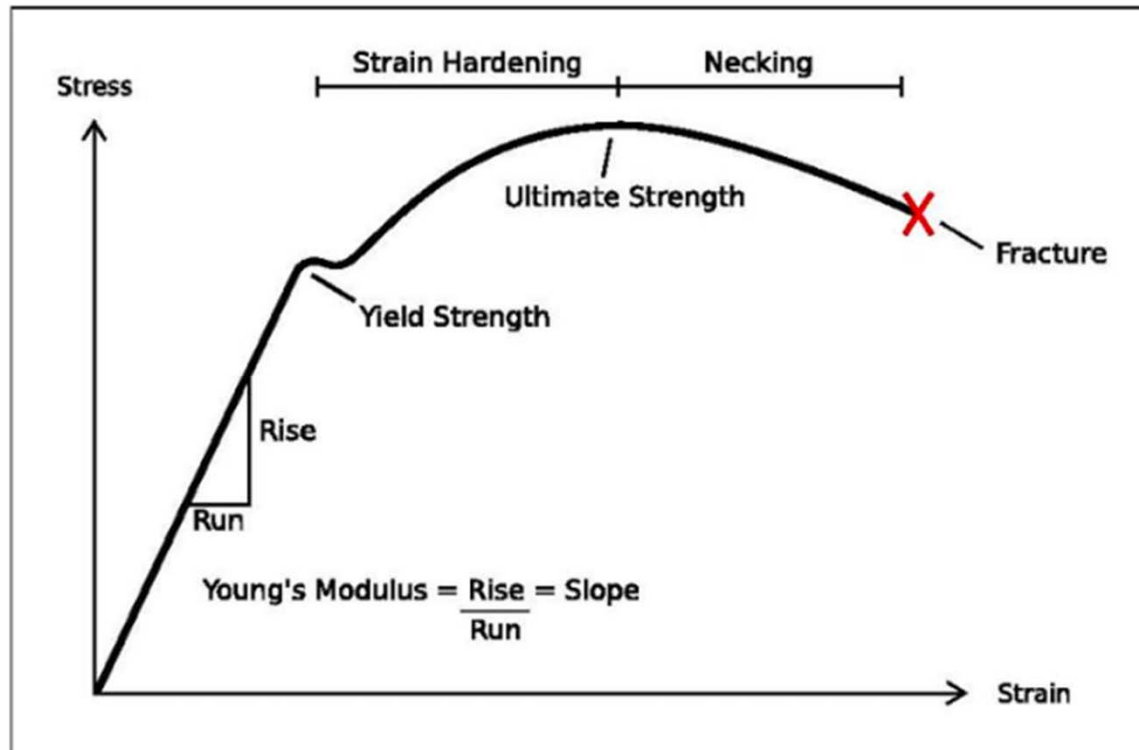


Table DS-1: Drill pipe grades.

Current Grades	Grade Code	Minimum Yield (psi)
E-75	E	75,000
X-95	X	95,000
G-105	G	105,000
S-135	S	135,000
Z-140*	Z	140,000
V-150*	V	150,000
U-165*	U	165,000

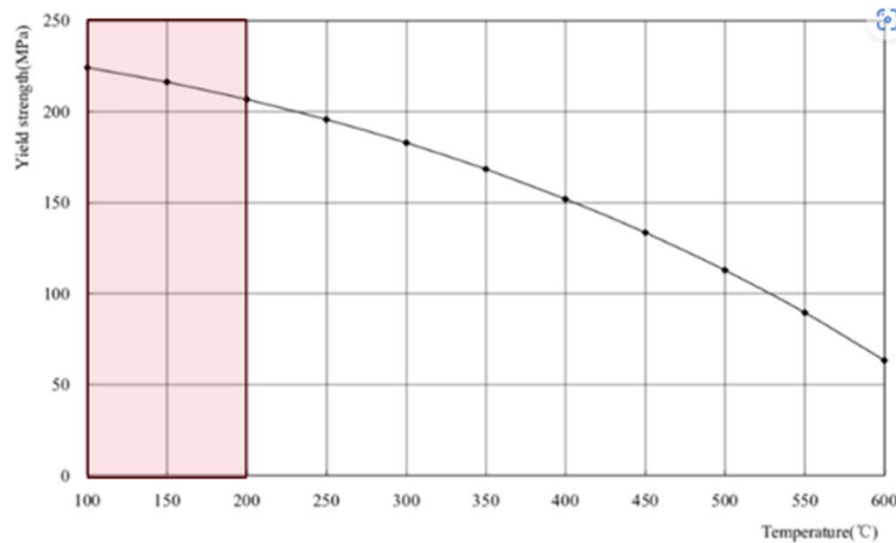
Drill pipe tubes are furnished in the following API length ranges:

- Range 1: 18-22 ft;
- Range 2: 27-30 ft;
- Range 3: 38-45 ft.

Environmental Effects

■ Temperature

- Increasing temperature will reduce the yield strength and endurance limit of steel.
- In the normal operating ranges of our tools of either 150 °C, 175 °C or 200°C a 12% to 15% reduction in Yield strength can occur



Environmental effects



- Corrosion
 - Is the deterioration of a material, usually a metal, by reaction with its environment.
 - It can be general, occurring across the entire exposed surface
 - It can be localized where only a small area of the surface is affected
 - It involves two electrochemical reactions, the dissolution of the metal and a reduction of a chemical species in the environment, in our case the drilling fluid.
- Sweet Corrosion
 - Carbon dioxide and water are necessary for this to occur, and the rate increases as the partial pressure of the CO₂ increases
- Sour Corrosion
 - The presence of Hydrogen Sulphide in the fluid is necessary for this to occur and leads to pitting and cracking of the steel





Thank you

Questions?



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